

Progesterone induces meiosis through two obligate co-receptors with PLA2 activity

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eLife Assessment

This **important** study provides **solid** evidence for a non-genomic action of progesterone in *Xenopus* oocyte activation. The findings demonstrate that two non-genomic progesterone receptors, ABHD2 and mPR β , function as a novel progesterone-stimulated phospholipase A2. The findings will be of broad interest to reproductive endocrinologists and physiologists.

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Abstract

The steroid hormone progesterone (P4) regulates multiple aspects of reproductive and metabolic physiology. Classical P4 signaling operates through nuclear receptors that regulate transcription. In addition, P4 signals through membrane P4 receptors (mPRs) in a rapid nongenomic modality. Despite the established physiological importance of P4 nongenomic signaling, the details of its signal transduction cascade remain elusive. Here, using *Xenopus* oocyte maturation as a well-established physiological readout of nongenomic P4 signaling, we identify the lipid hydrolase ABHD2 (α/β hydrolase domain-containing protein 2) as an essential mPR β co-receptor to trigger meiosis. We show using functional assays coupled to unbiased and targeted cell-based lipidomics that ABHD2 possesses a phospholipase A2 (PLA2) activity that requires mPR β . This PLA2 activity bifurcates P4 signaling by inducing clathrin-dependent endocytosis of mPR β , resulting in the production of lipid messengers that are G-protein coupled receptors agonists. Therefore, P4 drives meiosis by inducing an ABHD2 PLA2 activity that requires both mPR β and ABHD2 as obligate co-receptors.

Significance Statement

Nongenomic progesterone signaling is important for many physiological functions yet the details of its signaling remain elusive. Here we define the early signaling steps downstream of membrane progesterone receptor β (mPR β) during *Xenopus* oocyte meiosis. We show that progesterone requires two cell membrane receptors to work in unison to signal. The co-receptor complex possesses lipase activity that produces lipid messenger and induces receptor endocytosis to trigger meiosis progression. Our findings have broad physiological implications because nongenomic progesterone signaling operates in many tissues and regulates reproduction and metabolism.

Introduction

Progesterone (P4) signaling is critical for the regulation of female reproduction, sperm activation, neuronal function, and modulation of the immune system (1–3). The classical mode of P4 signaling is through nuclear receptors (nPRs) that act as transcription factors and modulate gene expression, resulting in cellular responses on a relatively slow time scale (4). In addition, P4 mediates rapid signal transduction that is independent of transcription and has thus been referred to as nongenomic. This signaling modality can be mediated by membrane progesterone receptors (mPRs) that are members of the progestin and AdipoQ receptors family (PAQR), and consist of 11 receptors with 5 being specific to P4: PAQR5 (mPR γ), PAQR6 (mPR δ), PAQR7 (mPR α), PAQR8 (mPR β) and PAQR9 (mPR ϵ) (5). Several lines of evidence support the importance of nongenomic mPR-dependent signaling, including results from studies of nPR knockout mouse lines, the speed of the signal transmission following P4 treatments, and the observed activity of membrane impermeant BSA-coupled P4 (3, 6–8). Currently, mPR-mediated signaling is recognized as an important regulator of many biological functions in the nervous and cardiovascular systems, female and male reproductive tissues, intestine, immune and cancer cells, and glucose homeostasis (3, 6, 9–11). Thus, mPRs are emerging as potential clinical targets for hypertension, reproductive disorders, and neurological diseases (12). However, their downstream signaling events are not well defined.

mPRs are integral membrane proteins with predicted 7-transmembrane domains, a cytoplasmic N-terminus and an extracellular C-terminus, which were first identified in fish ovaries almost 2 decades ago (5, 13, 14). This topology is opposite to that of G-protein coupled receptors (GPCRs), yet there is an abundance of evidence for G protein activation downstream of mPRs, primarily G α i and also G $\beta\gamma$ (15). However, there are conflicting reports as to whether mPRs signal through trimeric G-proteins or other modalities that remain to be defined (9, 16).

The anuran *Xenopus laevis* is a particularly suitable model to study P4 nongenomic signaling for two reasons: 1) the oocyte is transcriptionally silent, so signaling can only occur through the nongenomic arm; 2) P4 is known to trigger *Xenopus* oocyte maturation through the activation of mPR β (PAQR8) (14, 17). Fully grown vertebrate oocytes arrest at prophase I of meiosis for prolonged periods before undergoing maturation in preparation for fertilization (18). Importantly, P4 triggers re-entry into meiosis as well as a series of cytoplasmic differentiation steps that allow the egg to become fertilization-competent, able to activate in response to sperm entry, and to support early embryogenesis (19). Signaling downstream of mPR β ultimately leads to activation of maturation promoting factor (MPF), composed of Cdk1 and cyclin B, that serves as the primary kinase which triggers oocyte entry into M-phase (20). MPF is activated non-reversibly through two signaling cascades, Plk-Cdc25C and the MAPK cascade (Fig. 1A).

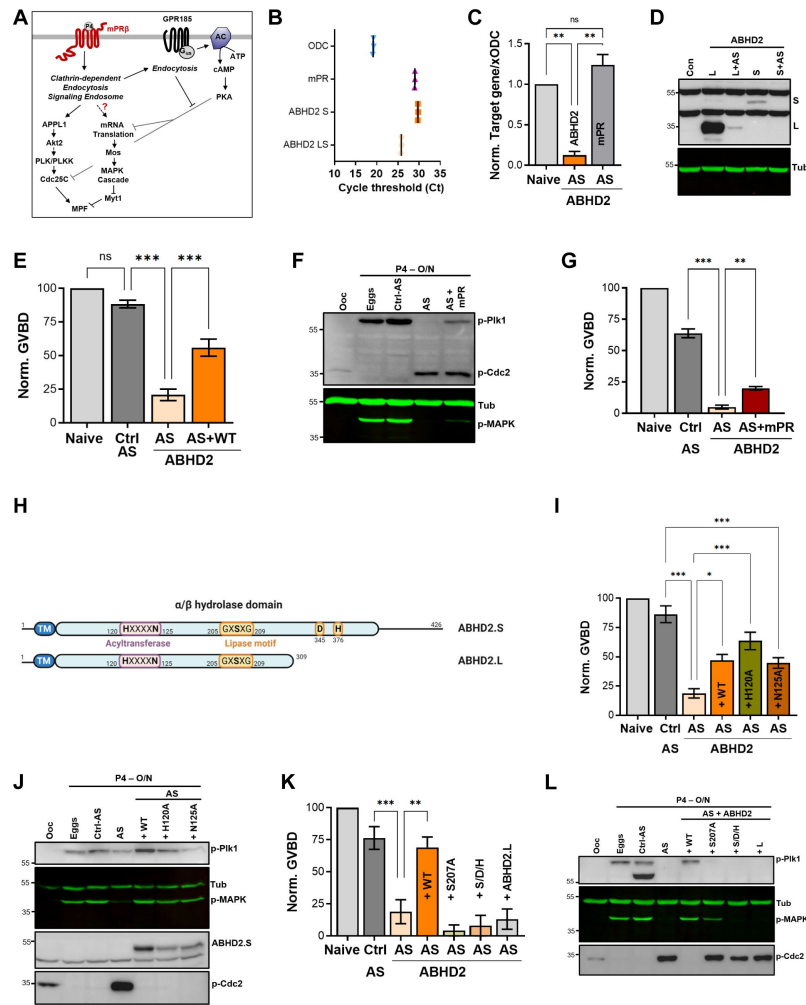


Figure 1.

ABHD2 is required for P4-induced oocyte maturation.

A. Signaling events downstream of mPR β after progesterone (P4), leading to oocyte maturation. The role of GPR185 is also depicted. **B.** mRNAs transcripts levels of mPR β , ABHD2.S, ABHD2.LS and *Xenopus* Ornithine decarboxylase (xODC) in oocytes using the Cycle threshold (Ct) generated from real time PCR. **C.** ABHD2 knockdown experiments. Oocytes were injected with specific ABHD2 antisense (AS) oligonucleotides and incubated at 18 °C for 24 hours. RNAs were prepared and analyzed by RT-PCR to determine the efficacy of ABHD2 knockdown as compared to naïve oocytes and mPR β RNAs levels. Data are expressed as relative RNAs levels of ABHD2 and mPR β mRNA after normalizing to xODC as a house keeping gene. **D.** Representative WB of ABHD2 in naïve and oocytes over-expressing ABHD2.L or ABHD2.S, following ABHD2 antisense (AS) oligonucleotide injection. Tubulin is shown as a loading control. **E.** Oocyte maturation following injection of control antisense (Ctrl AS) or ABHD2 antisense (AS) with or without overexpression of ABHD2.S (AS+WT) and normalized to P4-treated naïve oocytes condition (Naïve), (mean $\pm$ SEM; n = 7 independent female frogs, ordinary one-way ANOVA). **F.** Representative WB of MAPK, Plk1 and Cdc2 phosphorylation from untreated oocytes, eggs matured by overnight (O/N) treatment with P4, oocytes injected with control antisense (Ctrl AS) or ABHD2 antisense (AS) with or without overexpression of mPR β . Tubulin is shown as a loading control. **G.** Oocyte maturation in oocytes injected with control antisense (Ctrl AS) or ABHD2 antisense (AS) with or without overexpression of mPR β and normalized to P4-treated naïve oocytes (mean $\pm$ SEM; n = 3 independent female frogs, ordinary one-way ANOVA). **H.** Schematic representation of *Xenopus* ABHD2.S and .L domains, including the acyltransferase and lipase motifs. **I/K.** Oocyte maturation in oocytes injected with control antisense (Ctrl AS) or ABHD2 antisense (AS) with or without overexpression of ABHD2.S wild type (AS+WT) and the different ABHD2.S mutants as indicated, and normalized to P4-treated naïve oocytes (mean $\pm$ SEM; n = 3 independent female frogs, ordinary one-way ANOVA). **J/L.** Representative WB of MAPK, Plk1 and Cdc2 phosphorylation from untreated oocytes, eggs matured by overnight (O/N) with P4, oocytes injected with control antisense (Ctrl AS) or ABHD2 antisense (AS) with or without overexpression of the different ABHD2.S mutants as indicated on the panel. Tubulin is shown as a loading control.

In addition, oocyte meiotic arrest is maintained by high levels of cAMP-PKA through the action of a constitutively active GPCR, GPR185 (21 [↗](#), 22 [↗](#)). PKA blocks maturation by inhibiting both arms of the pathway: translation and Cdc25C (Fig. 1A [↗](#)) (23 [↗](#), 24 [↗](#)). Interestingly, GPR185 is internalized in response to P4, which inhibits its ability to activate adenylate cyclase and thus renders the oocyte more permissive for maturation (22 [↗](#)). Several studies have detected a drop in cAMP levels in response to P4 using ELISA (22 [↗](#), 25 [↗](#)–28 [↗](#)), but these findings could not be replicated using various reporters for cAMP and PKA at the single-cell level in real time (28 [↗](#)). Furthermore, the different sensors did not detect a global drop in cAMP in response to P4, and P4-dependent maturation proceeds even when cAMP is high (28 [↗](#)). This argues that P4 signals through parallel pathways that can overwhelm the cAMP-PKA-dependent inhibition.

We have previously shown that mPR β signals through APPL1 and AKT2 to activate Plk and MPF by forming a complex of activators within signaling endosomes (14 [↗](#)). Importantly, enrichment of mPR β within endosomes is sufficient to induce maturation in the absence of P4, arguing that a primary function of P4 is to induce clathrin-dependent endocytosis of mPR β (14 [↗](#)).

The lipase α/β hydrolase domain-containing protein 2 (ABHD2), an integral plasma membrane protein belonging to the ABHD family, was identified as a membrane progesterone receptor in human sperm (12 [↗](#), 29 [↗](#)) ABHD2 acts a monoacyl glycerol lipase (MAGL) that hydrolyses 2-arachidonoylglycerol (2-AG) forming arachidonic acid (AA) and glycerol and leads to sperm activation (29 [↗](#)). ABHD2 has also been implicated in follicular maturation in mammals (30 [↗](#)), vascular smooth muscle migration, and in pulmonary emphysema (31 [↗](#), 32 [↗](#)). In addition, ABHD2 was proposed to function as a triacylglycerol (TAG) lipase (33 [↗](#)), and shown to localize to the ER where it regulates Ca^{2+} release (34 [↗](#)).

Here we show that ABHD2 is required for the P4-induced release of oocyte meiotic arrest. Using untargeted lipidomic analyses, we detect broad downregulation of glycerophospholipid (GPL) and sphingolipid lipid metabolites in association with the enrichment of a key bioactive lipid messengers that include prostaglandins (PGs), lysophosphatidic acid (LPA), and potentially sphingosine-1-phosphate (S1P). Importantly, we show that ABHD2 acts as a PLA2 that requires mPR β to generate the aforementioned lipid messengers. The ABHD2 evoked PLA2 activity also triggers mPR β endocytosis, which as we have previously shown is sufficient to initiate oocyte maturation (14 [↗](#)). To our knowledge this is the first example of an α/β hydrolase that requires a heterologous co-receptor for activation.

Results

ABHD2 is required for P4-induced oocyte maturation

While reviewing the expression levels of various progesterone receptors in the *Xenopus* ovary, we noticed high levels of expression of ABHD2 in the oocyte and egg (Supplemental Fig. 1A). *Xenopus laevis* is tetraploid so it expresses two different versions of each gene, called S and L. There is significant sequence conservation between xABDH2.S and xABDH2.L with the human isoform, expect that xABDH2.L is truncated at position 309 (Supplemental Fig. 1B). Using primers specific to xABDH2.S and ones that amplify both the S and L isoforms, we confirmed that ABHD2 RNA is expressed in the oocyte at levels similar to those of mPR β RNA (Fig. 1B [↗](#)).

To test whether ABHD2 plays a role in oocyte maturation, we downregulated its expression by injecting anti-ABHD2 (L/S) oligos, which resulted in significant reduction of ABHD2 RNA levels without affecting the levels of mPR β RNA (Fig. 1C [↗](#)). We could not detect endogenous oocyte ABHD2 using Western blots (WB) most likely due to low expression levels, as we could readily detect overexpressed ABHD2 (Fig. 1D [↗](#)). We thus tested the efficacy of the antisense (AS) approach on overexpressed ABHD2 (either L or S). Figure 1D [↗](#) shows effective reduction of both

isoforms (**Fig. 1D**) within 24 hours after antisense injection, arguing for a short ABHD2 half-life on the order of hours. Interestingly, knocking down ABHD2 expression inhibited oocyte maturation in response to P4 (**Fig. 1E**), and was coupled to blocking Plk1, MAPK, and MPF activation (**Fig. 1F**).

This inhibition was significantly reversed by overexpression of ABHD2 (**Fig. 1E**), confirming that the specific knockdown of ABHD2 – and not off-target effects – mediated AS action. Overexpression of mPR β in ABHD2 antisense treated oocytes (Supplemental Fig. 1C) was less effective at rescuing oocyte maturation although it did have a significant effect (**Fig. 1G**), and partially rescued Plk1 and MAPK activation but not MPF (**Fig. 1F**). These data argue that both mPR β and ABHD2 are required to sufficiently activate the kinase cascades downstream of P4 to induce oocyte maturation.

ABHD2 lipase domain is required for P4-induced oocyte maturation

ABHD2 is a serine hydrolase that belongs to the α/β hydrolase family with a conserved lipid hydrolase domain GxSxG with residues D345 and H376 being important for lipase activity based on sequence homology (**Fig. 1H**). In addition, as with other members of the α/β hydrolase family, *Xenopus* ABHD2 has an acyltransferase domain that is close to the human HxxxD consensus but with Asn replacing the Asp residue (**Fig. 1H**). As ABHD2 has been shown to function as a P4-dependent hydrolase that is important for sperm activation (29), we wondered whether ABHD2 lipase activity is similarly important for oocyte maturation.

To evaluate this possibility, we replaced the conserved functionally relevant residues in the HxxxD acyltransferase domain motif (i.e., H120 and N125) with Ala and tested the mutants' ability to induce oocyte maturation in response to P4. Both the H120A and N125A mutants induced oocyte maturation to similar levels as those observed with WT ABHD2 in oocytes injected with ABHD2- AS (**Fig. 1I**), and this rescue was associated with effective expression of both mutants and activation of Plk1, MAPK, and MPF in eggs (**Fig. 1J**). These findings demonstrate that the ABHD2 acyltransferase motif is not required for maturation.

To test for the role of ABHD2 lipid hydrolase activity in oocyte maturation, we mutated all three catalytically relevant residues: S207 within the GxSxG motif, D345, and H376 to Ala (S/D/H mutant) or just S207 to Ala. Effective expression of either the S/D/H or S207A mutants (Supplemental Fig. 1D), did not induce maturation in oocytes where endogenous ABHD2 was knocked down (**Fig. 1K**). Neither mutant activated Plk1 or MPF, but expression of S207A led to low level MAPK activation (**Fig. 1L**) that was not associated with maturation (**Fig. 1K**). We further tested whether the shorter ABHD2.L (309 residues) is functional as it is missing the C-terminal end of the α/β hydrolase domain including both D345 and H376. Notably, ABHD2.L did not induce oocyte maturation in response to P4 (**Fig. 1K**) nor did it activate Plk1, MAPK, or MPF (**Fig. 1L**). These data show that all three residues involved in lipase activity (S207, D345, H376) are critical for ABHD2 P4-mediated oocyte maturation.

Lipidomics during oocyte maturation

The requirement for the lipase domain within ABHD2 for P4-dependent oocyte maturation suggests a potential role for lipid messengers downstream of P4 to release oocyte meiotic arrest. There is support for this idea in the literature from several early studies implicating lipid messengers in oocyte maturation, although there has been little consensus regarding specific pathways, lipases, or lipid mediators (35). To globally assess oocyte lipid profiles in response to P4, we performed unbiased mass-spectrometry based lipidomics and metabolomics (using the Metabolon CLP and HD4 platforms) at two time points after P4: 5 min to assess rapid changes in lipid abundances, and 30 min, a 'point of no return' where the majority of oocytes in the

population commit to maturation (22 [↗](#)). Furthermore, to be able to differentiate between pathways that are activated specifically through mPR β or ABHD2, we performed the profiling on oocytes following mPR β or ABHD2 knockdown.

The role of lipid signaling in oocyte maturation remains poorly defined, partly due to technical limitations in early studies, but also as a result of the complexity and transient nature of lipid signals (35 [↗](#)). An additional confounding factor in oocyte lipidomics is that oocytes in the population mature in an asynchronous fashion in response to P4. This asynchrony makes the identification of transient lipid messengers challenging. Finally, metabolomic profiles depend on environmental factors, basal metabolic rates, diet, and physical activity and are as such quite variable. We observe this variability as metabolotypes that are specific to individual frog donors in the principal component analysis in data from both the HD4 and CLP platforms (Supplemental Fig. 2A).

To minimize these technical and biological confounding factors, we performed the metabolomics analysis on groups of 10 oocytes at each time point and looked for relative changes in specific metabolites over time for each donor (Fig. 2A [↗](#)). That is the metabolic profile at T0 served as the baseline for each frog and we looked for changes in each metabolite at the 5- and 30-min time points after P4 addition. Furthermore, we averaged data from specific metabolites at the sub-pathway level to identify changes that are consistent among the 3 donor frogs tested. Data at the specific metabolite levels for both the CLP and HD4 platforms are presented in Supplementary Tables 1-6.

Lipidomics analyses on the CLP platform shows downregulation in response to P4 of multiple sphingolipid species, including ceramide, hexosylceramide, lactosylceramide, and sphingomyelin at both the 5 min and 30 min time points (Fig. 2B [↗](#) and Supplemental Tables 1-6). A similar decrease in multiple glycerophospholipid species, including phosphatidylethanolamine (PE), phosphatidylcholine (PC), lysoPE, lysoPC, as well as glycerides including monoacylglycerol (MAG), and diacylglycerol (DAG) is observed (Fig. 2B [↗](#) and Supplemental Tables 1-6). The HD4 platform analysis supports and extends these findings although at a lower level of granularity in terms of specific lipid species showing downregulation of multiple fatty acid species including arachidonic acid (AA) (Supplemental Fig. 2B). Note that most of these changes at the specific biochemical levels are relatively small (20-25% decrease) yet they gain relevance through their combined trend through multiple experiments at the levels of multiple metabolites within the same pathway (Supplemental Tables 1-6).

Importantly, knockdown of either ABHD2 or mPR β eliminated the downregulation of sphingolipids and glycerophospholipids in response to P4 at both the 5- and 30-min time points (Fig. 2B [↗](#) and Supplemental Fig. 2B). As oocytes with mPR β or ABHD2 knockdown do not mature in response to P4, the lipidomics data argue that the changes observed in the lipid metabolites are important for oocyte maturation; and that both mPR β and ABHD2 are required to initiate the lipase activities underlying these changes. The fact that many of those changes are observed early on after P4 addition (5 min) argues that they represent the initial signaling step downstream of P4 to commit the oocyte to meiosis.

We were initially puzzled by the global lipidomics profiles as they showed mostly downregulation of both GPL and sphingolipid species. If the cells are indeed in broad lipid catabolism mode, one would expect enrichment in some lipid end products. However, because oocyte maturation is asynchronous, changes in individual lipid species may be masked when several oocytes are pooled as in our lipidomics analyses. To get better insights into changes at the single oocyte level, we developed approaches (see Methods) to extract and analyze metabolites at the single oocyte level at the 30 min post-P4 time point. Single oocytes data showed a 3-fold increase in LPA and replicated the downregulation of GPL and sphingolipid species observed in the pooled analysis

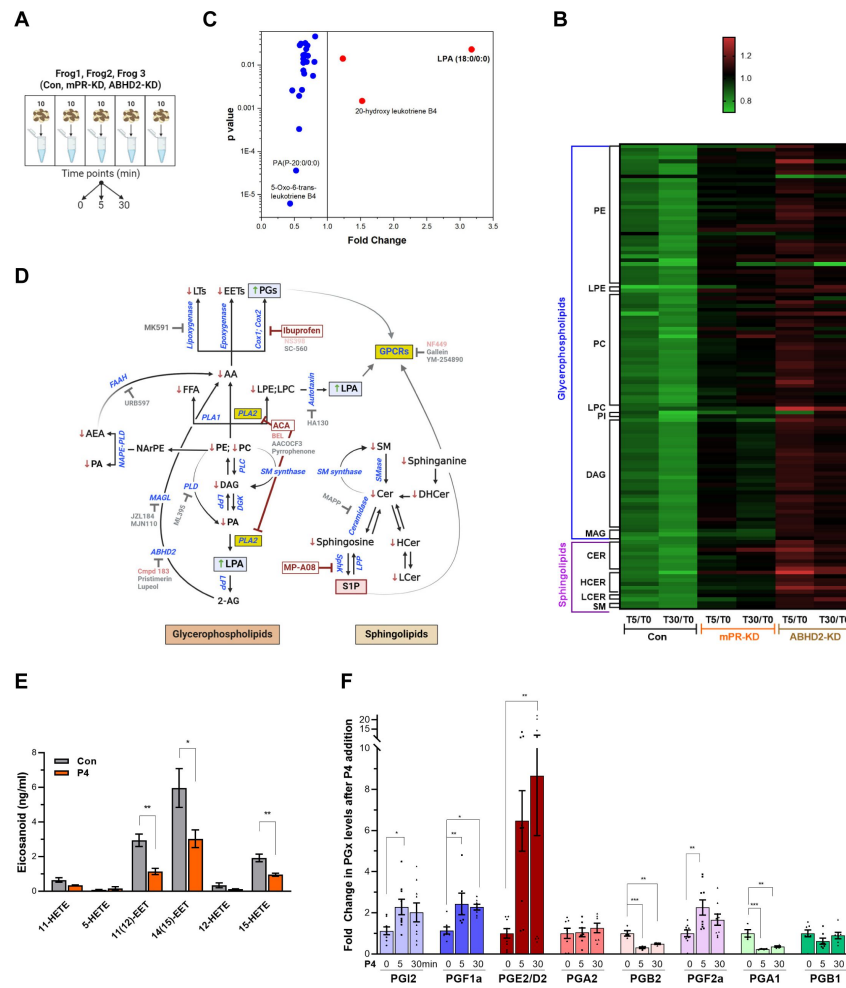


Figure 2.

Metabolomics and lipidomics **A. Summary of the experimental setup.** Naïve oocytes (Con) and oocytes injected with either mPR (mPR-KD) or ABHD2 antisense (ABHD2-KD) were treated for 5 or 30 min with P4. For each condition and time point, 5 replicates of 10 pooled oocytes each were collected. The experiment was repeated using 3 donor females. **B.** Heatmap of fold changes for metabolites that were changed significantly ($p < 0.05$) at either the 5 min (T5) or 30 min (T30) time points in response to P4 as compared to untreated oocytes (T5/T0 and T30/T0) for naïve (Con), mPR-KD, and ABHD2-KD oocytes. Metabolites are clustered at the levels of glycerophospholipids and then at the pathway level as follows: PE (Phosphatidylethanolamines), LPE (Lysophosphatidylethanolamines), PC (Phosphatidylcholines), LPC (Lysophosphatidylcholine), PI (Phosphatidylinositols), DAG (Diacylglycerols), MAG (Monoacylglycerols), CER (Ceramides), HCER (Hexosylceramides), LCER (Lactosylceramides), and SM (Sphingomyelins). Supplemental tables 1-4 show the means and p value for each ratio fold change. **C.** Distribution of metabolites that are reduced (blue) or increased (red) significantly in single naïve oocytes 30 min after P4. Fold change and p values were calculated from 10 individual oocytes at each time point. The raw data is listed in Supplemental Table 7. **D.** Summary of the changes in sphingolipids and glycerophospholipids after progesterone treatment. Increase (upward green arrow) or decrease (downward red arrow) in metabolites levels is noted. Tested chemical inhibitors are also shown. Strong inhibitors are indicated in red, weak inhibitors in pink, and drugs that do not inhibit oocyte maturation in gray. Generated using Biorender. **E.** Levels of EETs and HETEs before and after P4 treatment in single oocytes. Naïve oocytes were incubated with either ethanol or P4 10^{-5} M for 30 min. 20 single oocytes per condition were collected and used for analysis. 5-oxoETE and 8(9)-EET were detected in 1 or 2 samples respectively, so they were not included in the statistical analyses although both were lower following P4 treatment. **F.** Levels of prostaglandins before (0 min) and after P4 treatment at 5 min and 30 min time points in naïve oocytes. Per each condition, 10 replicates were collected containing 10 pooled oocytes each. The following eicosanoids were not detected in either group: 6kPGF1 α , PGF2 α , PGE2, TXB2, PGD2, PGA2, PGJ2, 12-HHTre, 11-dehydro TXB2, LTB4, LTC4, LTD4, LTE4, 20-hydroxy LTB4, 20-carboxy LTB4, 5(6)-DiHETEs, LXA4, LXB4, 5(6)-EET, 5(6)-DiHET, 8(9)-DiHET, 11(12)-DiHET, 14(15)-DHET, 20-HETE. Similar results were obtained from individual oocyte samples (see Supplemental Table 7). Data are normalized to the PG levels at time zero. Example of the raw abundance of individual PG species is shown in Supplemental Fig. 2C.

(**Fig. 2C** [↗](#)). The lipidomics changes observed in response to P4 are summarized in **Figure 2D** [↗](#) along both the GPL and sphingolipid pathways by an arrow up (red) or down (green) next to each detected lipid species.

Some end products of GPL and SL metabolism such as eicosanoids and sphingosine 1 phosphate (S1P) are not well suited to the standard extraction protocols used for our global metabolomics studies. Therefore, to enrich for these end products we performed targeted extraction and MS analyses against validated standards for eicosanoids and S1P, which would be the most likely end metabolites enriched based on the downregulation profile of other metabolites throughout the GPL and SL pathways (**Fig. 2D** [↗](#)). AA is a precursor for eicosanoids, which are produced through three main enzymatic pathways: 1) cyclooxygenases (cox1/cox2) action results in the formation of prostaglandins; 2) lipoxygenases produce leukotrienes; and 3) cytochrome P450 enzymes, ω -hydrolases and epoxigenases, yield hydroxyeicosatetraenoic acids (HETEs) and epoxyeicosatrienoic acids (EETs), respectively (**Fig. 2D** [↗](#)).

We could not detect any leukotrienes in the oocyte samples despite testing for multiple species (LTB4, LTC4, LTD4, LTE4, 20-hydroxy LTB4, 20-carboxy LTB4) but could observe a downregulation in their precursor hydroxy-eicosatetraenoic acids (HETE) (**Fig. 2E** [↗](#)). We did however detect a decrease in LTB4 derivatives in the single oocyte MS analyses (**Fig. 2B** [↗](#)). We also observed decreases in two epoxyeicosatrienoids (EET) from single oocyte samples (20 oocytes at 30 min post P4) (**Fig. 2E** [↗](#)). In contrast to leukotrienes and lipoxins, the trend in prostaglandins (PG) was the opposite with increases observed in multiple species, including PGI2, PGF1a, PGE2/PGD2, and PGF2a (**Fig. 2F** [↗](#)). **Figure 2E** [↗](#) shows the relative changes of PG species from different donor females. To illustrate the basal abundance of the different PGs species relative to each other, Supplemental Fig. 2C shows the raw abundance of the different PGs relative to each other. Collectively these targeted metabolomics profiles show that AA is preferentially metabolized by cyclooxygenases at the expense of lipoxygenases/epoxigenases to produce PGs in response to P4 (**Fig. 2D** [↗](#)).

From the metabolomics analyses of the SL pathway, sphingosine-1-phosphate (S1P) is predicted to increase in response to P4, as all other SL metabolites decrease in response to P4 (**Fig. 2D** [↗](#)). We therefore directly tested S1P levels in the oocyte in response to P4 at 5 and 30 min but did not detect any significant change (Supplemental Fig. 2D). This implies that either S1P does not change in response to P4, or alternatively that the S1P produced in response to P4 diffuses out of the oocyte. This is possible as S1P is membrane permeant and acts on cell surface receptors from the extracellular side ([36](#) [↗](#)). We tested this possibility directly below by assessing the role of S1P receptors in oocyte maturation.

Pharmacological validation of the metabolomics findings

The metabolomics findings from extensive analyses using multiple platforms and extraction procedures to enrich and detect specific lipid species argue for an important role for lipid messengers in inducing oocyte maturation. This is further supported by the fact that when either mPR β or ABHD2 are knocked down the lipid changes disappear. However, these metabolomics studies are correlative and do not address cause-effect relationships. To test whether activation of specific lipases is required for oocyte maturation, we used pharmacological tools to inhibit specific enzymes predicted to be important for maturation based on our metabolomic analyses (**Fig. 2D** [↗](#)).

Along the GPL metabolic arm, the lipidomics data show downregulation of upstream metabolites with enrichment of LPA and prostaglandins (PGs) (**Fig. 2** [↗](#)). This argues for a central role for PLA2 activity as it would produce AA as a precursor for PGs and LPC/LPE as precursors for LPA. To test for a role for PLA2 in maturation, we used a broad-spectrum inhibitor, ACA, as PLA2s represent a large family of enzymes (up to 16 groups) with disparate activation modes, Ca²⁺-dependency, and subcellular localization (secreted or cytosolic) ([37](#) [↗](#), [38](#) [↗](#)). ACA completely inhibited maturation

with an IC_{50} of 3.8×10^{-6} M (**Fig. 3A**). ACA treatment blocked Plk1, MAPK, and MPF activation explaining the maturation inhibition (**Fig. 3B**). Given the robust inhibition with ACA, we tested other PLA2 inhibitors, including AACOF3, bromoenol lactone (BEL), and pyrrophenone to get insights into the PLA2 isoform involved. BEL inhibited oocyte maturation with an IC_{50} of 3.9×10^{-5} M (**Fig. 3A**), by mainly blocking the Plk1 pathway (**Fig. 3C**). The BEL IC_{50} was about 5-fold higher than its reported inhibitory potency of 8×10^{-6} M (**Table 1**). The other two PLA2 inhibitors were ineffective (**Fig. 3A**). It is difficult to pinpoint a particular PLA2 isoform based on the isoform specific profile of these inhibitors (see <https://www.sigmaaldrich.com/QA/en/technical-documents/technical-article/protein-biology/protein-expression/phospholipase-a2>), arguing that the oocyte's PLA2 activity required for maturation does not match known PLA2 isoforms. Nonetheless, the inhibitors data support a central role for PLA2 in inducing oocyte maturation downstream of P4. In line with this, pre-treatment of oocytes with mastoparan, a peptide toxin from wasp venom known to activate PLA2, enhanced oocyte maturation by ~50% in response to low P4 (**Fig. 3D**).

AA could still be produced through the action of fatty acid amide hydrolase (FAAH) independently of PLA2 activation, and PLD activation produces phosphatidic acid (PA) which is a precursor of LPA (**Fig. 2F**). Inhibition of either FAAH or PLD had no effect on oocyte maturation (Supplemental Fig. 3A), arguing that the precursors for LPA and PGs are produced primarily through PLA2 activation in response to P4 in the oocyte.

Given the inhibitory effect of ABHD2 knockdown on oocyte maturation, we tested the effect of two plant triterpenoids known to inhibit ABHD2 and MAGLs, pristimerin and lupeol (39), as well as MAGL chemical inhibitors JZL184 and MJN110. None of these inhibitors had a significant inhibitory effect on oocyte maturation at high P4 (Supplemental Fig. 3B). However, because pristimerin showed a trend toward inhibition at the highest concentration, we tested its effect at low P4 concentration where it inhibited oocyte maturation but required a concentration of 10^{-5} M (Supplemental Fig. 3C), which is two orders of magnitude higher than its documented IC_{50} in other systems (39). These results argue against an important role for a monoacylglycerol enzymatic activity, whether ABHD2-dependent or not, in inducing oocyte maturation. We further tested a reported specific ABHD2 inhibitor, compound 183 (10^{-4} M), which inhibited oocyte maturation at low and high P4 concentrations (Supplemental Fig. 3D), arguing for a role for ABHD2 in oocyte maturation.

We next focused on enzymes that generate PGs and LPA as the enriched lipid metabolites in response to P4 (**Fig. 2D**). LPA can be produced through the action of PLA2 on phosphatidic acid (PA) or through autotaxin metabolism of LPE/LPC (**Fig. 2D**). Blocking autotaxin activity with HA130 repressed oocyte maturation with an IC_{50} of 4.2×10^{-5} M in the presence of high P4 (**Fig. 3E**). We therefore tested the effects of HA130 at low P4, where it significantly inhibited maturation by ~80% (**Fig. 3F**). However, this inhibition of maturation required high concentrations of HA130 -at least 3 orders of magnitude higher than the reported HA130 IC_{50} , arguing against a major role for the autotaxin pathway in generating LPA in response to P4, and rather supporting LPA production primarily through PLA2 activity.

We then tested the effect of inhibiting cyclooxygenases responsible for PGs production on oocyte maturation. Inhibition of Cox enzymes with ibuprofen effectively blocked maturation with an IC_{50} of 2.7×10^{-4} M consistent with its IC_{50} against Cox2 (3.7×10^{-4} M) (**Fig. 3E**, **Table 1**). As expected from its effect on maturation, ibuprofen treatment blocked both Plk1 and MAPK activation (**Fig. 3G**). Specific inhibition of Cox2 using NS398 blocked maturation with an IC_{50} of 10^{-4} M (**Fig. 3E**), which is higher than its documented IC_{50} against Cox2 (3.8×10^{-6} M) (**Table 1**). Consistently, NS398 at high concentration had a more pronounced inhibitory effect on oocyte maturation (~85%) at limiting P4 concentrations (**Fig. 3H**). The Cox1 specific inhibitor SC-560 had no effect on oocyte maturation (**Fig. 3E**). Collectively these data argue for a role for Cox2 in supporting maturation but not for Cox1.

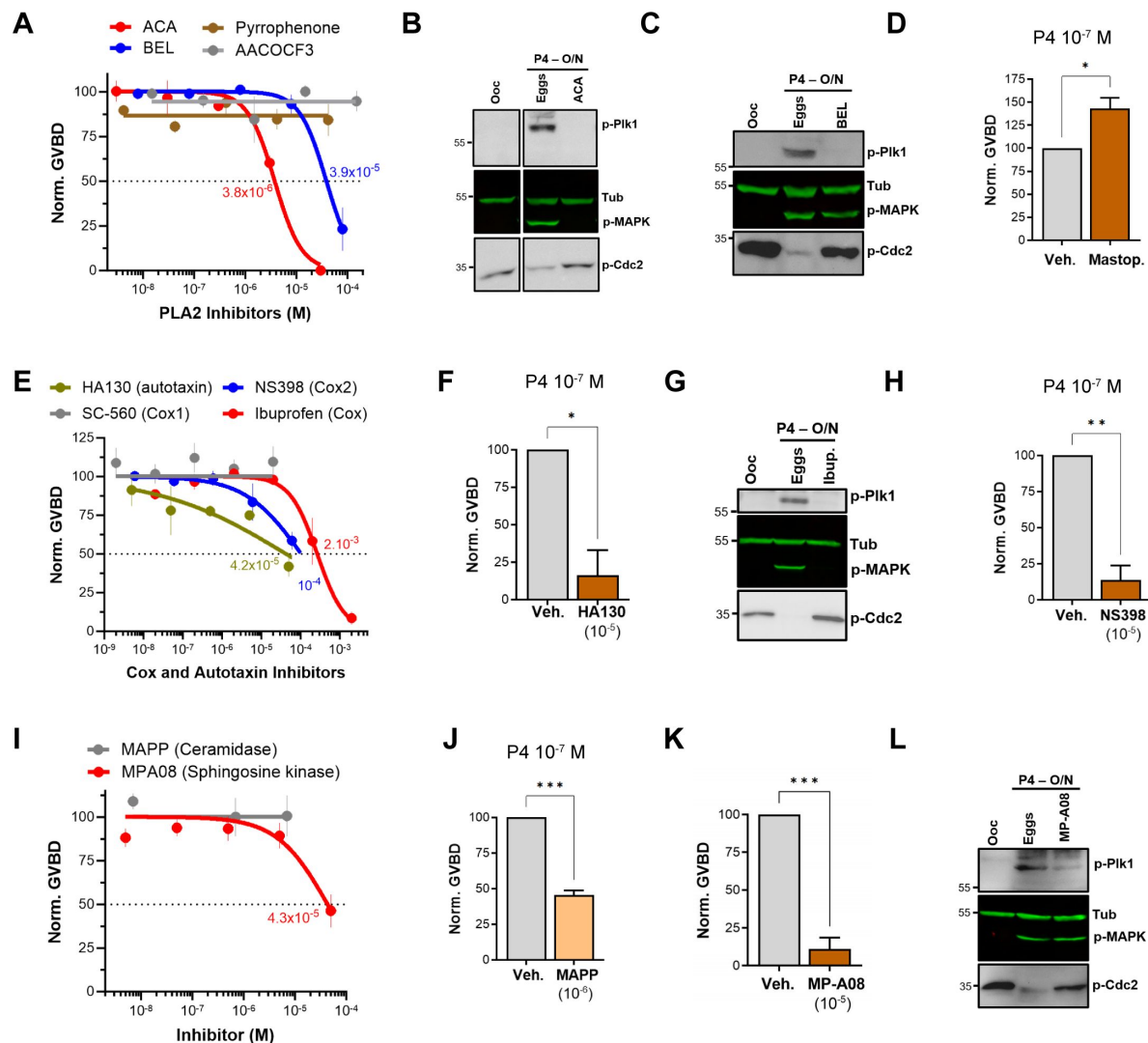


Figure 3.

Pharmacological validation of the metabolomics findings. A/E/I.

Dose response of inhibition of oocyte maturation for the drugs tested. Oocytes were pre-treated for 2 hours with vehicle or with increasing concentrations of the indicated drugs, followed by overnight treatment with P4 at 10⁻⁵M. IC₅₀ was calculated using a nonlinear regression fit (mean ± SEM; n = 3 independent female frogs). **D/F/H/J/K.** Drug effect on oocytes maturation at low P4 concentration. Oocytes were pre-treated for 2 hours with the vehicle or with the highest drug concentration from the dose response, followed by overnight treatment with P4 at 10⁻⁷M. Oocyte maturation was normalized to control oocytes (treated with vehicle) (mean ± SEM; n = 3 independent female frogs for each chemical compound experiment, unpaired t-test). **B/C/G/L.** Representative WB of MAPK, Plk1 and Cdc2 phosphorylation from untreated oocytes, oocytes pretreated with vehicle or the indicated drug for 2 hours then matured overnight (O/N) with P4 (eggs). Tubulin is shown as a loading control.

Inhibitor	Target	IC ₅₀ ± SEM Oocyte Maturation (M)	IC ₅₀ (M) Representative Study	Ratio (Ooc/Lit)
ACA	PLA2	3.8E-6 ± 5.8E-7	5E-6 (54)	0.76
BEL	PLA2	3.9E-5 ± 8E-6	8E-6 (55)	4.9
MP-A08	Sphingosine Kinase 1	4.3E-5 ± 1E-5	2.7E-5 (56)	1.6
HA130	Autotaxin	4.2E-5 ± 4.8E-5	2.8E-8 (57)	1,522
Ibuprofen	Cox-2	2.7E-4 ± 6E-5	3.7E-4 (58)	0.73
NS398	Cox-2	1E-4 ± 5E-5	3.8E-6 (59)	26.3
NF-449	GαS	5.2E-5 ± 3E-5	7.9E-6 (60)	6.5

Table 1.

Pharmacological inhibitors potency

Interestingly, inhibition of lipoxygenases using MK591 potentiated maturation by ~20% in the presence of high P4 (Supplemental Fig. 3E), and by 100% at low P4 (Supplemental Fig. 3F). Leukotrienes are decreased in response to P4 (**Fig. 2**), so presumably inhibition of lipoxygenases diverts AA metabolism toward cyclooxygenase hydrolysis thus increasing PGs production, which would explain the observed effect on oocyte maturation.

S1P signaling supports oocyte maturation

The lipidomics data show downregulation of most metabolites along the sphingolipid arm of the metabolic pathway with a presumed enrichment of S1P (**Fig. 2D**). We were however unable to detect an increase of S1P in the oocyte in response to P4 even when using a targeted mass spectrometry-based assay for S1P (Supplemental Fig. 2D). This may be because S1P is readily secreted and signals extracellularly. However, activation of ceramidases could be involved in the sphingolipid changes. mPRs share sequence and structural homology with adiponectin receptors, which have been shown to possess ceramidase activity that requires Zn in the active site that acts on the ceramide amide bond (40). We have previously shown that an mPR β mutant with all 4 zinc coordinating residues mutated to Ala to abrogate Zn binding (H129,D146,H281,H285A) is functional in inducing oocyte maturation (14). This argues that mPR β does not require ceramidase activity to release oocyte meiotic arrest. To confirm this finding, we mutated Ser125 in the conserved SxxxH motif in mPR β , which matches the ceramidase domain in PAQR and AdipoQ receptors (41, 42). The mPR β -S125A mutant expresses and traffics normally to the cell membrane to similar levels as WT mPR β (Supplemental Fig. 4A). Furthermore, it signals effectively downstream of progesterone as its overexpression (Supplemental Fig. 4B) rescues mPR β knockdown with similar efficiency as WT mPR β (Supplemental Fig. 4C), and activates Plk1, MAPK, and MPF to induce oocyte maturation (Supplemental Fig. 4D). Collectively these results argue against a role for mPR β -dependent ceramidase activity in inducing oocyte maturation. Hence, changes in sphingolipids might be a consequence of the alterations in other lipid metabolic pathways especially that the metabolism of GPLs is linked to that of sphingolipids through the activity of SM synthase.

If indeed S1P is the end product of sphingolipid metabolism in response to P4, then one would predict activation of sphingosine kinases and concurrently ceramidases to support the metabolic flux. We tested this hypothesis using broad spectrum sphingosine kinase and ceramidase inhibitors. Inhibition of ceramidase activity using D-e-MAPP had little effect on oocyte maturation at high P4 (10^{-5} M) (**Fig. 3I**). In contrast, at limiting P4 (10^{-7} M) MAPP inhibited oocyte maturation by ~55% compared to untreated oocytes (**Fig. 3J**). Note that at low P4 (10^{-7} M) oocyte maturation is not complete as only the most primed oocytes respond, and this response varies from frog to frog (10-50% of the maturation observed with high P4). Inhibition of sphingosine kinases using MP-A08 inhibited oocyte maturation with an IC₅₀ of 4.3×10^{-5} M at high P4 (**Fig. 3I** and **Table 1**). At low P4, MP-A08 blocked maturation by 87.5% (**Fig. 3K**), supporting a role for S1P in oocyte maturation. MP-A08 treatment primarily blocks Plk1 activation and to a lesser extent MAPK (**Fig. 3L**).

To assess the role of S1P signaling in releasing oocyte meiotic arrest, we targeted S1P receptor (S1PR) isoforms based on their expression in the ovary. Of the 5 S1PR isoforms, only S1PR2 and S1PR3 are expressed in the ovary with higher expression of S1PR3 (43). We first confirmed the expression profile in oocytes using RT-PCR (**Fig. 4A**). S1PR3 expression levels were higher than S1PR2 (**Fig. 4A**). We confirmed the efficiency of the S1PR2 primers in the spleen where S1PR2 was shown to be highly expressed (43). We then focused on S1PR3 as the primary S1PR in the oocyte and knocked down its expression using antisense oligos. We validated S1PR3 knockdown at the RNA (**Fig. 4B**) and protein (**Fig. 4C**) levels. S1PR3 knockdown blocked oocyte maturation induced by either P4 or the mPR specific agonist OD-02 by ~50% (**Fig. 4D**).

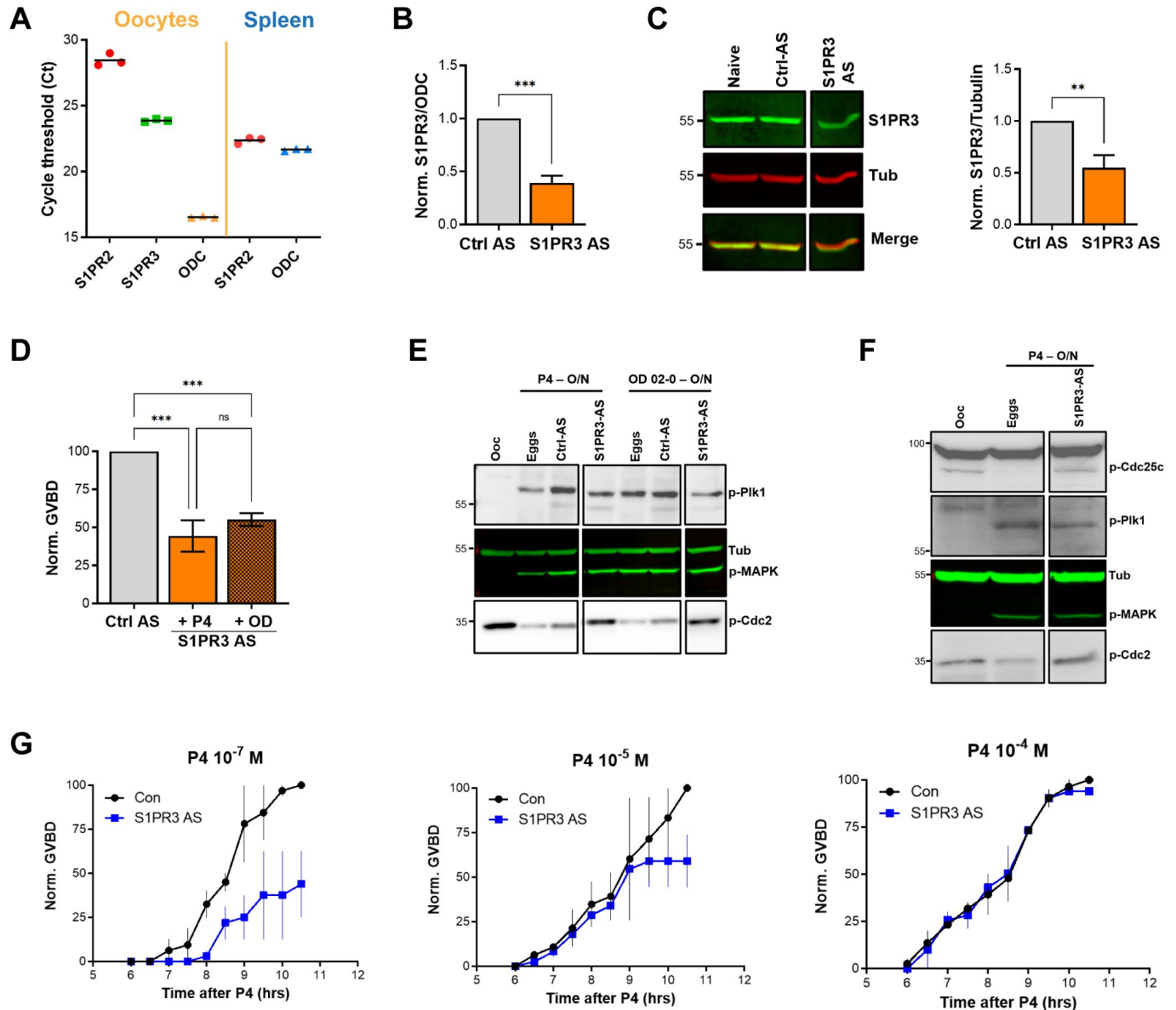


Figure 4.

S1P signaling and oocyte maturation.

A. mRNAs levels of S1PR2, S1PR3, and the house keeping gene Ornithine decarboxylase (ODC) in oocytes and spleen measured as the Cycle threshold (Ct) from real time PCR. **B-C.** S1PR3 knockdown. Oocytes were injected with control antisense (Ctrl AS) or specific S1PR3 antisense oligonucleotides and incubated at 18 °C for 24 hours. RNAs and proteins extracts were prepared and analyzed by RT-PCR and WB to determine the efficacy of S1PR3 knockdown as compared to control oocytes (Ctrl AS). **B.** Histogram showing the relative RNAs levels of S1PR3 mRNA to xODC. **C.** Representative WB (left panel) and normalized quantification (right panel) comparing S1PR3 protein levels between naïve, Ctrl AS and S1PR3 AS injected oocytes. Tubulin is shown as a loading control (mean $\pm$ SEM; n = 6 independent female frogs). **D.** Oocyte maturation following injection of S1PR3 antisense, normalized to P4 or Org OD 02-0 (OD)-treated oocytes injected with control antisense (Ctrl AS) (mean $\pm$ SEM; n = 7 independent female frogs, ordinary one-way ANOVA). **E/F.** Representative WBs of MAPK, Plk1 and Cdc2 (E), as well as CDC25C (F) phosphorylation on untreated oocytes, P4 or OD matured eggs (O/N)D, oocytes injected with control antisense (Ctrl AS) or S1PR3 antisense (S1PR3 AS) and treated O/N with P4 or OD. Tubulin is shown as a loading control. **G.** GVBD-time course after treatment with P4 at the indicated concentrations in oocytes injected with water (Con) or with S1PR3 antisense (S1PR3 AS) (mean $\pm$ SEM; n = 2 independent female frogs).

Interestingly, S1PR3 knockdown did not affect P4-mediated activation of MAPK and Plk1 (Fig. 4E), yet P4 was unable to dephosphorylate and activate MPF (Fig. 4E). Plk1 induces maturation by activating the dual-specificity phosphatase Cdc25C, which dephosphorylates Cdc2 to activate MPF and is considered a rate limiting step in MPF activation (44). We therefore tested the effects of S1PR3 knockdown on Cdc25C activation and show that despite the induction of both Plk1 and MAPK, Cdc25C was not activated in response to P4 in S1PR3 knockdown cells (Fig. 4F). We followed Cdc25C activation by assessing its dephosphorylation at Ser287 (Ser216 in human), which is required for its activation (Fig. 4F) (44). The partial oocyte maturation block in S1PR3 KD oocytes argues for a modulatory role for S1P signaling. We tested this hypothesis by assessing the requirement for S1PR3 at increasing P4 dosages that would incrementally induce the multiple arms of the maturation signaling pathways. S1PR3 knockdown oocytes matured with significantly less efficiency and with a slower time course at low P4 (10^{-7} M) (Fig. 4G), but interestingly this effect was gradually lost with increased P4 concentrations (10^{-5} and 10^{-4} M) (Fig. 4G). These data argue that the S1P/S1PR3 pathway modulates oocyte maturation, but that by itself it is not essential for maturation as it can be bypassed when other pathways are hyperactivated at high P4 concentrations. This is consistent with the production of multiple lipid messengers to induce maturation through several GPCR pathways.

Role of G-protein coupled receptors (GPCRs) in mediating lipid messenger action

All the enriched lipid metabolites detected from our lipidomics analyses - PGs, LPA, and possibly S1P - act through G-protein coupled receptors (GPCRs). This raises the intriguing possibility that P4 expands its signaling through the production of lipid messengers that act on several GPCRs to mediate the multiple cellular changes that need to occur concurrently during oocyte maturation. This is an attractive possibility as GPCRs have been implicated in *Xenopus* oocyte maturation for decades without conclusive evidence for their involvement, and they have been shown to mediate mPR nongenomic signaling. Therefore, we tested the effects of broad inhibitors of trimeric G-proteins. Blocking Gas using NF-449 partially inhibited maturation ($IC_{50} 5 \times 10^{-5}$ M) (Supplemental Fig. 4E), whereas blocking either G $\beta\gamma$ or Gq/11 with gallein and YM-254890 respectively was inefficient at blocking oocyte maturation (Supplemental Fig. 4E). Interestingly, inhibiting Gas with NF449 primarily blocked Plk1 activation to inhibit MPF and oocyte maturation, and to a lesser extent the MAPK pathway (Supplemental Fig. 4F). At low P4 concentration the inhibitory effect of NF-449 was still observed as well as inhibition by gallein but not YM-254890 (Supplemental Fig. 4G). This argues for a supporting role for G $\beta\gamma$ signaling as well in oocyte maturation. These results support a role for Gas signaling and potentially G $\beta\gamma$ downstream of P4 through lipid intermediates that require PLA2, Cox2, and SphK activities.

ABHD2 is a PLA2 that requires mPR β as a co-receptor

Both the metabolomics and pharmacological data strongly implicate PLA2 activation downstream of P4 to induce maturation. In addition, several lines of evidence argue that this PLA2 activity requires ABHD2: 1) Induction of the signaling cascades downstream of P4 requires both ABHD2 and PLA2; 2) the lipidomics changes downstream of P4 are dependent on both activities; 3) mutations in the ABHD2 lipase domain abolish its ability to induce maturation; 4) ABHD2 has been characterized as a lipase acting on either MAG or TAG (29, 33); and 5) an ABHD2 specific inhibitor (compound 183) inhibits maturation. Collectively, these findings suggest that ABHD2 is associated with a PLA2 activity that is induced in response to P4 to release oocyte meiotic arrest.

To test this hypothesis, we translated ABHD2 and mPR β *in vitro* in rabbit reticulocyte lysates and assessed whether they possess PLA2 activity. Expression of ABHD2 or mPR β alone was not associated with any increase in PLA2 activity whether in the presence or absence of P4 (Fig. 5A). Interestingly though, co-expression of both mPR β and ABHD2 produced a PLA2 activity, but only in the presence of P4 (Fig. 5A). This despite our inability to detect either mPR or ABHD2 expression by Western blots from the reticulocyte lysates reactions. These results argue that both

mPR β and ABHD2 are required for the P4-dependent PLA2 activity, but they do not address which receptor is the catalytic subunit. However, our functional experiment with the ABHD2 hydrolase domain mutants (S/D/H and S207) strongly argues that ABHD2 is the PLA2 catalytic subunit.

To directly confirm this prediction, we co-expressed mPR β with either ABHD2 WT or the S/D/H mutant in reticulocyte lysates and tested their PLA2 activity. Mutating the ABHD2 lipid hydrolase domain (S/D/H mutant) resulted in loss of PLA2 activity when the protein was expressed with mPR β in the presence of P4 (**Fig. 5B**). Taken together, these results argue that ABHD2 has PLA2 catalytic activity but only in the presence of mPR β .

To cross validate our findings from the reticulocyte experiments, we tested in vitro and cell-based expression systems to increase protein yield. We settled on the coupled in vitro transcription/translation tobacco lysates system (ALICE®) that produces high protein yields and targets expressed proteins to microsomes directly using a melittin signal peptide. This is an advantage for our proteins of interest as they are both integral membrane proteins. The tobacco lysates expression system produced significantly higher yields of both mPR and ABHD2 expression where both proteins were readily detected on Western blots (**Fig. 5C**). Similar to the reticulocyte system, tobacco lysates expressing ABHD2 or mPR β alone were not associated increased PLA2 activity whether in the presence or absence of P4 (**Fig. 5D**). Of note tobacco lysates have endogenous PLA2 activity, which was inhibited by expression of mPR β alone (**Fig. 5D**). The PLA2 activity data from tobacco lysates are reported as the rate of increase over time (**Fig. 5D**). Co-expression of mPR β and ABHD2 in the tobacco lysates produced significant PLA2 activity that was not only higher than that detected from the reticulocyte lysates but also stable over extended periods of time allowing recordings for over 2 hours (**Fig. 5D**). Interestingly in this case however the PLA2 activity of the co-receptor complex was independent of P4 (**Fig. 5D**). Both the reticulocyte and tobacco expression system show that mPR β is a necessary partner for ABHD2 to mediate its PLA2 activities. However, the reticulocyte co-expression results argue that mPR β and ABHD2 need P4 to assemble and form a complex to be able to mediate the ABHD2 PLA2 activity. In contrast, when the two proteins are expressed at a high levels and more important targeted specifically to the same microsomal compartment their PLA2 activity becomes P4 independent. We were thus interested in testing whether the two receptors interact in vivo. We expressed mPR β -GFP in oocytes and tested whether it forms a complex with ABHD2. Immunoprecipitation of mPR β -GFP pulls down ABHD2 at rest (**Fig. 5E**), and importantly this interaction is significantly enhanced (1.9 $\pm$ 0.2-fold, $p=0.002$) following P4 treatment (**Fig. 5E**). This argues that P4 enhances the assembly of the co-receptor complex. Furthermore, co-expression and pull-down experiments show that the non-functional ABHD2 mutants: S/D/H and ABHD2.L interact with mPR β (Supplemental Fig. 4H), indicating that the interaction between ABHD2 and mPR β does not require a functional lipase domain.

ABHD2 and PLA2 activity are required for mPR β endocytosis and signaling

We have previously shown that mPR β induces oocyte maturation at the level of the signaling endosome following its clathrin-dependent endocytosis (14). Importantly, mPR β endocytosis is sufficient to induce maturation in the absence of P4. Since changes in plasma membrane lipid composition are known to modulate endocytosis and membrane components activities, we suspected a role for ABHD2/PLA2 in mPR β endocytosis in response to P4. Therefore, we quantified mPR β endocytosis following ABHD2 knockdown. P4 leads to enrichment of mPR β positive intracellular vesicles, and this enrichment requires ABHD2 as it was lost in ABHD2 knockdown oocytes (**Fig. 6A**). PLA2 activity was also critical for P4 induced mPR β endocytosis, which was completely blocked in ACA treated oocytes (**Fig. 6B**). Interestingly, ACA also inhibited basal endocytosis of mPR β in the absence of P4, which is apparent as lower vesicular mPR β in oocytes treated with ACA (**Fig. 6B**, no P4). This argues that both ABHD2 and PLA2 are required for mPR β endocytosis and by extension its signaling in response to P4.

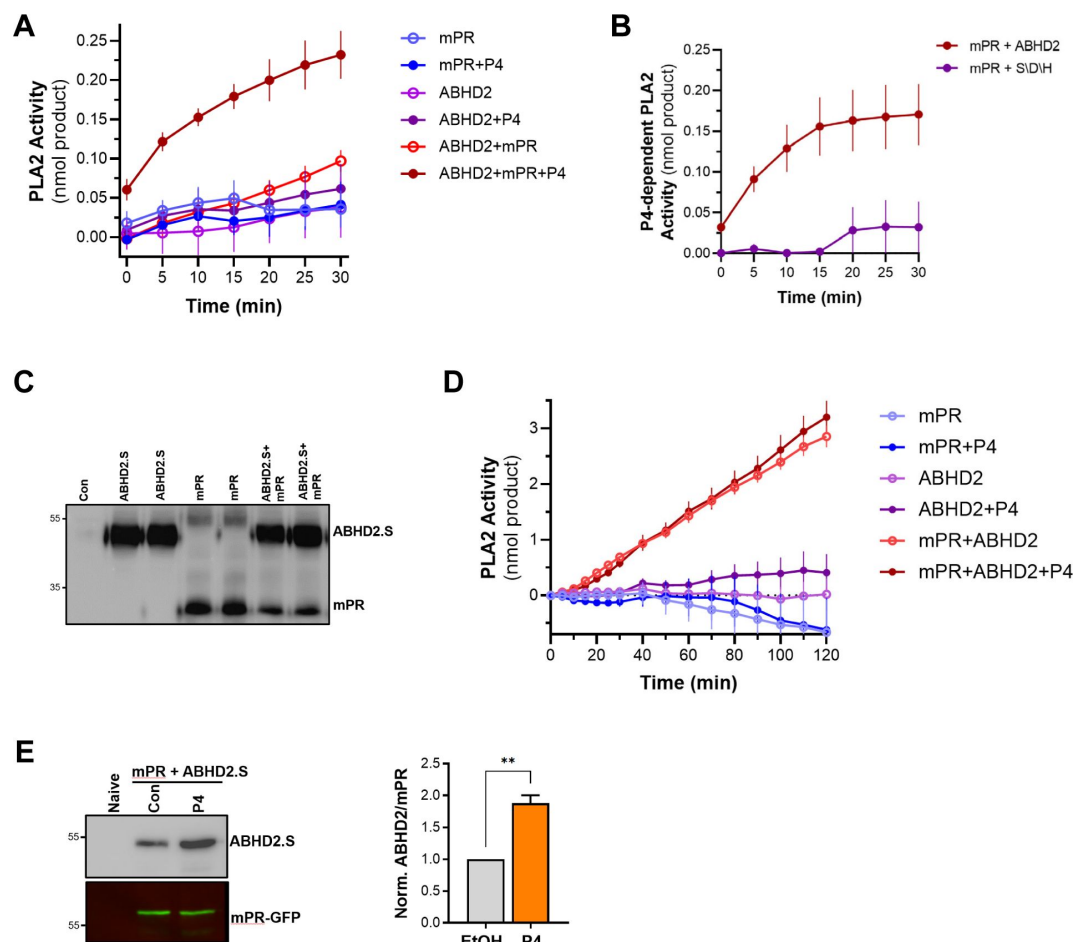


Figure 5.

PLA2 activity of the mPR-ABHD2 co-receptor complex.

A. Time-dependent PLA2 activity from reticulocyte lysates expressing mPR, ABHD2.S, or ABHD2.S+mPR in the presence of ethanol as vehicle or P4 10^{-5} M (mean $\pm$ SEM; $n = 3$). **B.** Time course of P4-dependent PLA2 activity in reticulocyte lysates expressing mPR with either wild-type ABHD2 (ABHD2) or the ABHD2 S207A/D345A/H376A mutant (S/D/H). P4-dependent PLA2 is plotted as the difference in activity in the presence and absence of P4 (Mean $\pm$ SEM; $n = 3$). **C.** Example WB probed with anti-His antibody from tobacco lysates (ALiCE®) alone (Con) and lysates expressing mPR or ABHD2 alone or both proteins in duplicated as indicated. **D.** Time-dependent PLA2 activity from ALiCE® lysates overexpressing mPR, ABHD2.S, or ABHD2.S+mPR in the presence of the vehicle ethanol or P4 10^{-5} M (mean $\pm$ SEM; $n = 3$). Data are plotted as the rate of production of the lysothiophospholipid product from the beginning of the experiment (0 min) with the rate of the lysates alone subtracted. **E.** Representative immunoprecipitation (IP) WB and quantification of mPR-GFP from oocyte lysates from un-injected oocytes (Naive) or oocytes over-expressing mPR-GFP and ABHD2.S treated for 40 min with Ethanol (EtOH) or P4. *Left panel*, the representative WB membrane is probed for ABHD2 and GFP (mean $\pm$ SEM; $n = 3$ independent female frogs, unpaired t-test).

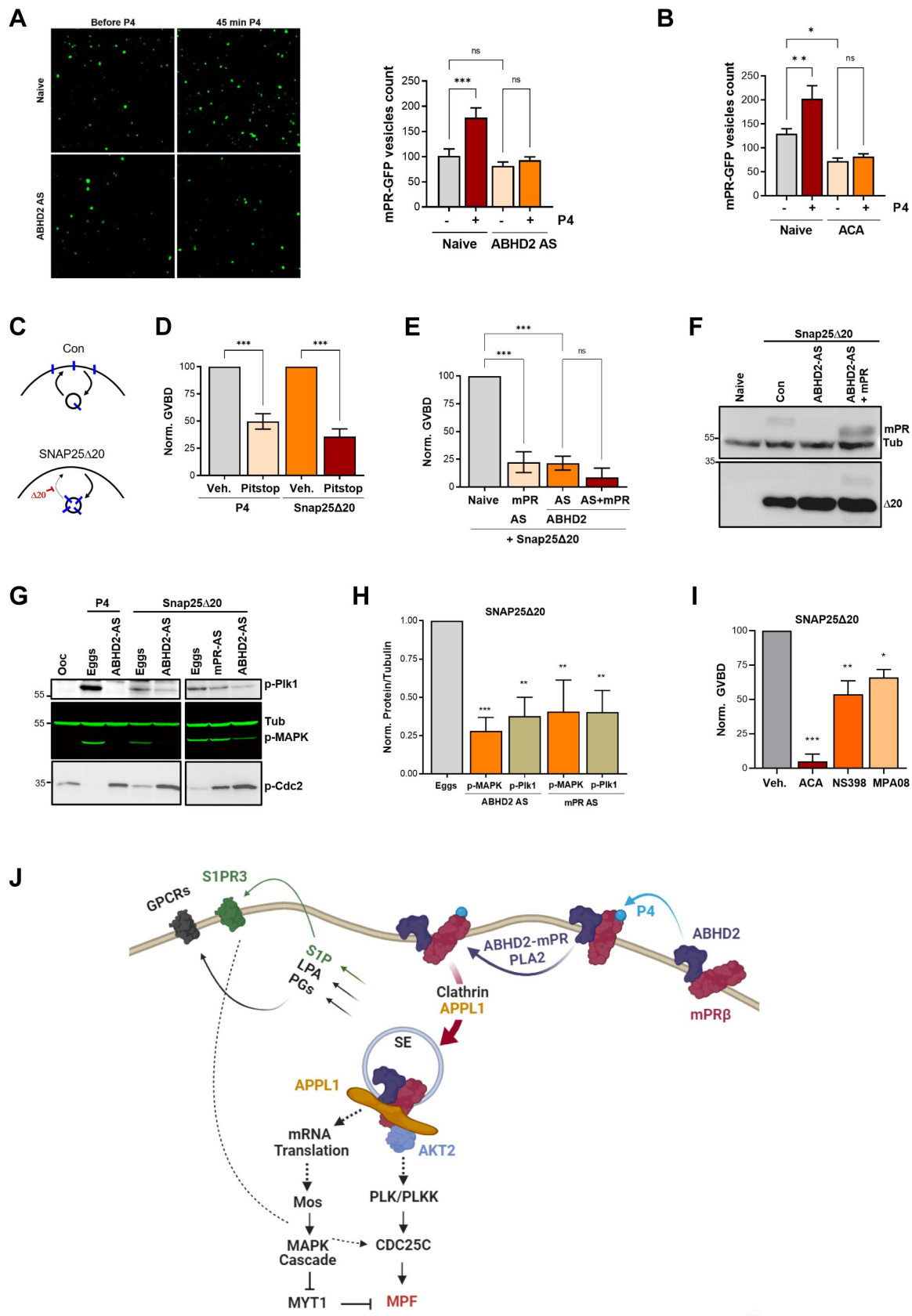


Figure 6.

ABHD2 and PLA2 activity are required for mPR β endocytosis and signaling.

A. Effect of ABHD2 knockdown on mPR-P4 mediated endocytosis. Oocytes were injected with mPR-GFP RNA (Naïve) and 48 hours later either left untreated or injected with ABHD2 antisense (ABHD2 AS). The following day oocytes were imaged, before and 45 min after P4 treatment. *Left panel*, representative confocal image of mPR-GFP positive vesicles in naïve and ABHD2 AS oocytes, before and after P4. *Right panel*, histogram of mPR-GFP positive vesicles count before and after P4 (mean $\pm$ SEM; n = 20 to 21 oocytes per condition, from 3 independent female frogs, ordinary one-way ANOVA). **B.** PLA2 inhibition blocks P4 mediated endocytosis. Vesicle count from oocytes expressing mPR-GFP for 72 hours were pre-treated with vehicle (naïve) or ACA for 2 hours, followed by imaging, before and 45 min after P4 treatment (mean $\pm$ SEM; n = 16 to 18 oocytes per condition, from 2 independent female frogs, ordinary one-way ANOVA). **C.** Cartoon depicting the role of SNAP25 Δ 20 in blocking exocytosis. **D.** SNAP25 Δ 20-induced oocyte maturation requires clathrin-dependent endocytosis. Naïve oocytes were pre-treated with vehicle or Pitstop (10^{-5} M), followed by overnight treatment with P4 or SNAP25 Δ 20-mRNA injection. Oocyte maturation in P4 or SNAP25 Δ 20 injected oocytes normalized to the vehicle condition (mean $\pm$ SEM; n = 4 independent female frogs, ordinary one-way ANOVA). **E.** SNAP25 Δ 20- induced maturation requires ABHD2. Oocytes were injected with mPR antisense (mPR AS) or ABHD2 antisense (AS) with or without mPR β mRNA (AS+mPR). 48 hours later, oocytes were injected with mRNA to overexpress SNAP25 Δ 20. The following day, oocyte maturation was measured in mPR AS, ABHD2 AS and ABHD2 AS+mPR oocytes normalized to naïve oocytes injected with SNAP25 Δ 20 (mean $\pm$ SEM; n = 3 independent female frogs, ordinary one-way ANOVA). **F.** Representative WB of mPR and SNAP25 Δ 20 proteins expression in naïve, ABHD2 AS and ABHD2 AS+mPR oocytes. Tubulin is shown as a loading control. **G.** Representative WB of MAPK, Plk1 and Cdc2 phosphorylation from untreated oocytes, P4 matured eggs, or SNAP25 Δ 20 mRNA injection, and oocytes injected with mPR (mPR AS) or ABHD2 antisense (ABHD2 AS) and treated O/N with P4 or SNAP25 Δ 20 mRNA injection. Tubulin is shown as a loading control. **H.** Quantification of p-Plk as the ratio of p-PLK/Tubulin and p-MAPK as the ratio of p-MAPK/Tubulin normalized to the ratios in naïve eggs (mean $\pm$ SEM; n = 4 independent female frogs, ordinary one-way ANOVA). **I.** Oocyte maturation in oocytes pretreated for 2 hours with ACA, NS398 and MP-A08 followed by SNAP25 Δ 20-mRNA injection and normalized to GVBD in oocytes pre-treated with vehicle followed by SNAP25 Δ 20-mRNA injection (Veh.) (mean $\pm$ SEM; n = 3 independent female frogs, ordinary one-way ANOVA). **J.** Model of the signaling cascade triggered in response to P4 (generated using Biorender).

We were therefore interested in assessing the subcellular distribution of ABHD2 during oocyte maturation to determine whether it is also internalized. We thus tagged it at the N- or C-terminus with mCherry. Unfortunately, both mCh-tagged ABHD2s were not functional as they did not rescue ABHD2 knockdown (Supplemental Fig. 4I) despite efficient expression and binding to mPR (Supplemental Fig. 4H). This prevented us from imaging the trafficking and distribution of ABHD2 during oocyte maturation.

The requirement for ABHD2 to mediate P4-dependent mPR β endocytosis (Fig. 6A) suggests that this may be the primary function of ABHD2 in response to P4. Under that scenario, P4 activates ABHD2 enzymatic activity which induces mPR β endocytosis leading to oocyte maturation. To test this hypothesis, we were interested in inducing mPR β endocytosis in the absence of P4. We have previously shown that a dominant negative SNAP25 mutant missing the last 20 residues (SNAP25 Δ 20), effectively blocks exocytosis in the oocyte (Fig. 6C); leading to mPR β enrichment intracellularly and inducing maturation in the absence of P4 (14, 45). Similar to P4, SNAP25 Δ 20- induced oocyte maturation requires clathrin-dependent endocytosis as it was blocked by clathrin blocker pitstop (Fig. 6D). As expected, SNAP25 Δ 20-induced oocyte maturation requires mPR β as it was inhibited in oocytes where mPR β was knocked down (Fig. 6E). Surprisingly, SNAP25 Δ 20- induced maturation also requires ABHD2, since oocytes where ABHD2 is knocked down were unable to mature in response to SNAP25 Δ 20 (Fig. 6E), despite high expression of SNAP25 Δ 20 (Fig. 6F). We further tested whether mPR β overexpression allows ABHD2 KD oocytes to mature in response to SNAP25 Δ 20, but this was not the case (Fig. 6E). This

argues that even in the presence of excess mPR β , its forced internalization using SNAP25 Δ 20 (**Fig. 6C**) in the absence of ABHD2 is not sufficient to induce maturation. Therefore, ABHD2 activity is not only required for mPR β internalization but it is also required at the level of the signaling endosome following mPR β endocytosis to induce maturation.

Like P4, SNAP25 Δ 20 activates Plk1, MAPK, and MPF (**Fig. 6G**) and this induction was significantly inhibited following knockdown of either ABHD2 or mPR β (**Fig. 6G** and **6H**), showing that both proteins signal through the canonical kinase cascades to induce maturation following internalization even in the absence of P4. Furthermore, in addition to the PLA2 inhibitor ACA, inhibition of Cox2, or sphingosine kinase partially blocked SNAP25 Δ 20 induced maturation with ACA being the most effective (**Fig. 6I**), similar to what we observe with P4. ACA blocks basal mPR β recycling (**Fig. 6B**), and as such would inhibit its internalization in response to SNAP25 Δ 20 expression. However, the Cox2 (NS398) and the SphK (MPA08) inhibitors should not interfere with mPR β internalization in response to SNAP25 Δ 20, although experimentally it is difficult to ascertain this. Collectively these results argue that PLA2 activity is required for mPR β internalization and that furthermore both Cox2 and SphK activities are important to modulate oocyte maturation following mPR β internalization.

Discussion

mPR-dependent nongenomic signaling is an important regulator of many physiological processes in female and male reproduction, cardiovascular, neuroendocrine, neurological and immune function (3, 6, 9, 12). This raises interest in mPRs as potential therapeutic targets for hypertension and other cardiovascular diseases, reproductive disorders, neurological diseases, and cancer (12). Therefore, understanding mPR signaling is important to assess their physiological and pathological contributions. In this study, we used oocyte maturation in the frog as a well-established model for P4 nongenomic signaling.

Oocyte maturation prepares the egg for fertilization and endows it with the ability to activate and initiate embryonic development (19). As such it represents the initial step in multicellular organismal development that precedes fertilization. It is therefore not surprising that oocyte maturation involves multiple signaling cascades to mediate both the completion of meiosis -the so-called nuclear maturation- to generate a haploid gamete, and the extensive cellular differentiation of the oocyte that is needed to support development, and includes protein synthesis, membrane remodeling, and Ca²⁺ signaling differentiation that supports the block of polyspermy and resumption of meiosis at fertilization (19).

In *Xenopus* oocytes, the release of the long-term meiotic arrest can be triggered by P4. In fact, *Xenopus* oocyte maturation is one of the most studied examples of nongenomic P4 signaling and represents a well characterized system to define the signaling cascade downstream of P4. It is well established that P4 in the oocyte activates two parallel and interdependent kinase cascades that ultimately culminate in MPF activation (20). Despite many studies over the past decades, the earliest signaling steps downstream of mPR have remained elusive. An interesting recent development shows that clathrin-dependent endocytosis of mPR β is necessary and sufficient for its signaling and the induction of oocyte maturation even in the absence of P4 (14). It therefore appears that the signaling endosome acts as a hub to activate the multiple pathways required to release meiotic arrest, resume meiosis, and prepare the egg for fertilization.

An untested assumption is that P4 induces oocyte maturation primarily through mPR β since its knockdown or anti-mPR β antibodies block maturation (14, 17). However, the oocyte expresses other P4 receptors including ABHD2 that may contribute to P4 signaling. Here we show that knockdown of ABHD2 blocks oocyte maturation and that this is rescued effectively by ABHD2 overexpression but not by mPR β overexpression. ABHD2 contains a lipase and an acyltransferase

motif. The lipase domain is required for oocyte maturation but not the acyltransferase domain. Based on this finding, we undertook an unbiased lipidomics approach to better define metabolic alterations driven by ABHD2 lipase activity. P4 resulted in a broad decrease in glycerophospholipid and sphingolipid species, with the enrichment of few lipid messenger end products, namely prostaglandins, LPA and S1P (**Fig. 6j** [46](#)). This is intriguing as all these lipid messengers act through GPCRs. We in fact validate S1P action through the S1P receptor, which is a GPCR, to support oocyte maturation. These findings argue that P4 activates ABHD2 lipase activity to generate lipid messengers that in turn stimulate various GPCRs that regulate the multiple aspects of oocyte maturation. This would nicely explain the standing controversy in the field as to whether mPRs are GPCRs or not. Should the findings from the oocyte hold in other systems, it would argue that P4 activates lipid catabolism to generate lipid messenger that then act through GPCRs. Thus, nongenomic P4 signaling would involve both non-GPCR and GPCR signaling modalities in series.

An intriguing finding of the current study is that the PLA2 activity of ABHD2 is activated through its interaction with mPR β (**Fig. 6j** [46](#)). Although ABHD2 and mPR β interact at rest, this interaction is enhanced following P4 treatment. This is the first report showing that ABHD2 possesses PLA2 activity and that this PLA2 activity requires interaction with mPR β . The PLA2 family is large and complex with over 50 members that have been classified based on enzymatic activity, co-factor dependence, subcellular distribution, and structure ([37](#) [46](#)); and include the secreted, cytosolic, Ca²⁺-independent, lysosomal, platelet-activating factor acyl hydrolase, and some members of the α/β hydrolase domain (ABHD) family ([47](#)).

ABHD2 has been shown to act as a 2-AG lipase to activate sperm ([29](#) [39](#)). Interestingly, we find no evidence for such an activity in response to P4 based on our lipidomics analyses in the oocyte. This raises the interesting possibility that ABHD2 lipase specificity may be regulated by protein-protein interactions in a cell specific fashion, but this remains to be tested.

In addition to the generation of lipid messengers that branch out P4 signaling at its onset into multiple signaling cascades through GPCRs, the ABHD2 PLA2 activity hydrolyzes phospholipids to generate AA that is then metabolized to PGs through the activity of Cox2. The hydrolysis of membrane phospholipids generates lysophospholipids with a single acyl chain that tend to have a conical shape. Such conical phospholipids are prone to inducing membrane curvature ([48](#)) as the latter is determined by the size of the lipid headgroups and hydrophobic tails ([49](#)). Conical lipids have been shown to play important roles in membrane fusion events both *in vitro* and *in vivo* ([50](#)–[52](#)). It is therefore tempting to postulate that enrichment in lysophospholipids following ABHD2- PLA2 activation generates spontaneous membrane curvature, which can be sensed by the clathrin endocytic machinery to induce mPR β endocytosis and initiate signaling and maturation (**Fig. 6j** [46](#)).

Collectively our findings define the earliest steps in nongenomic P4 signaling to release the oocyte meiotic arrest and prepare the egg for fertilization (**Fig. 6j** [46](#)). Furthermore, they show for the first time that two heterologous receptors previously implicated in nongenomic signaling, mPR β and ABHD2, function as coreceptors to induce PLA2 activity that is required for both endocytosis and further downstream signaling. Nongenomic P4 signaling is widespread and mediates reproductive, neurological, and other functions. Therefore, these findings are likely to have broad implications throughout biology.

Materials and methods

Reagents and Primers

A list of the antibodies, siRNA, chemicals and other reagents used is provided in Supplemental Table 7. Primers and sense/antisense oligos used are listed in Supplemental Table 8. Details of the different clones used is provided in Supplemental Data.

Xenopus laevis oocytes

All animal procedures and protocols were performed in accordance with the University of Weill Cornell Medicine-Qatar Institutional Animal Care and Use Committee. Stage VI *Xenopus laevis* oocytes were obtained as previously described (53). Oocytes were maintained in L-15 medium solution (Sigma-Aldrich Cat# L4386) supplemented with HEPES (Sigma-Aldrich Cat# H4034), 0.1% (v/v) of penicillin/streptomycin stock solution (ThermoFisher Scientific Cat# 15140-122) and 0.1% (v/v) of gentamycin (EMD Millipore Cat# 345814-1GM) at pH 7.6. The oocytes were used 24 to 72 hours after harvesting and injected with RNAs or sense/antisense oligos and kept at 18°C for 1-2 days to allow for protein expression or knockdown. After treatment with progesterone at 10^{-7} M, 10^{-5} M or 10^{-4} M (as indicated), GVBD was detected visually by the appearance of a white spot at the animal pole. For the inhibitor's studies, oocytes were preincubated for 2 hours with different concentration of the inhibitors, followed by overnight incubation with P4 at 10^{-7} M or 10^{-5} M concentration. Co-immunoprecipitation, Western blot, and oocyte imaging are described in supplemental data.

Metabolomic analyses

Untargeted metabolomics were performed on single oocytes or a group of 10 oocytes per condition as described in Supplemental Data. For measurements of metabolites in a group of 10 oocytes, 3 experiments using 3 independent female frogs were performed. Each experiment was conducted in 5 sample replicates per condition per frog. Metabolites were analyzed on both the Metabolon HD4 and CLP platforms.

Tandem LC MSMS was used to measure prostaglandins and S1P from 10 oocytes using 10 replicates per condition as described in Supplemental Data. Levels of Eicosanoids, EETs and HETEs before and 30 min after P4 treatment were measured in 20 single oocytes per condition by Cayman chemicals as described in Supplemental Data.

PLA2 assay

RNAs for mPR and ABHD2.S wt or ABHD2.S S/D/H mutants were used to generate recombinant proteins of mPR, ABHD2.S and mPR/ABHD2.S, using the rabbit reticulocytes lysates Nuclease-treated kit and the transcendTM tRNA from Promega, as per the manufacturer's instruction. These different reactions were used to measure phospholipase A2 (PLA2) activity following manufacturer's procedures (Abcam, Cambridge, UK), in the presence of ethanol or P4 (10^{-5} M). No RNA control without substrate were used as background. PLA2 activity was corrected on reticulocytes without any RNA.

We used a coupled in vitro transcription/translation kit (ALiCE[®] Cell-Free Protein Synthesis System, Sigma Aldrich) that allows the expression of difficult-to-produce proteins, such as membrane proteins, in microsomes using the pALICE02.His vector, which adds a melittin signal peptide to translocate the proteins into microsomes. To subclone mPR β into pALICE02 vector, pSGEM-mPR was PCR amplified using the following primers: 5' CTACCATGGCAATGACTACCGCAATCCTTGA-3' (Forward) 5'-CTAGGTACCTCAGTGATGGTGATGGTGATGAAGTTCTTTTCTGGCCAACT-3' (Reverse). The resulting PCR product was cut with NcoI and KpnI, gel purified, and ligated into NcoI/KpnI of pALICE02. To subclone ABHD2 in pALICE02, pSGEM-ABHD2 was PCR amplified using the following primers: 5'-ACTGATATCATGGATGCGATAGTGGAACCC-3'(Forward) and 5'-CTAGGTACCTCAGTGATGGTGATGGTGATGTTATGGTCAGACTCAGCAGCC-3'(Reverse). The resulting

PCR product was cut with EcoRV and KpnI, gel purified, and ligated into pALICE02 cut with NcoI and Klenow treated to produce a blunt end then cut with KpnI. All constructs were verified by DNA sequencing and by analytical endonuclease restriction enzyme digestion. The pALICE02 vector allows for the expression of C-terminally His-tagged mPR β and ABHD2. Briefly, DNA was added to the pALICE lysates, followed by incubation at 25°C for 48 hours with constant shaking at 700 rpm. The expression of ABHD2 and mPR β were confirmed by western blot, before proceeding to the PLA2 assay.

Statistics

Data are presented as mean $\pm$ SEM. Each set of experiments was at least repeated 3 times. Groups were compared using the Prism 9 software (GraphPad) using the statistical tests indicated in the figure legend. Statistical significance is indicated by p values (ns, not significant; *** $p \leq 0.001$; ** $p \leq 0.01$; * $p \leq 0.05$).

Acknowledgements

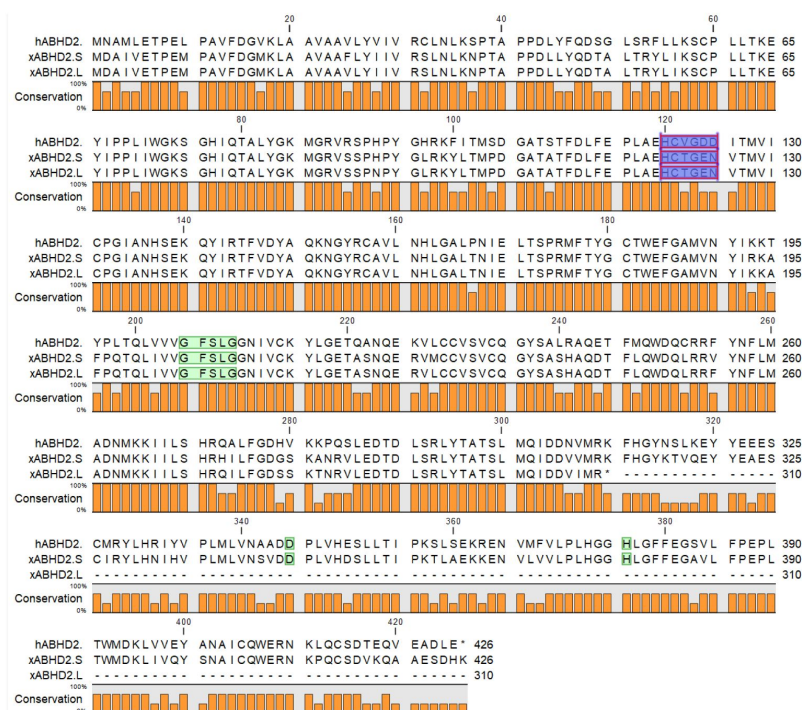
We are grateful to the Bioinformatics and Virtual Metabolomics Core at Weill Cornell Medicine Qatar for their support in the metabolomics and lipidomic studies. We also thank the Vivarium and Microscopy Cores at WCMQ for their support in multiple experiments. This work as well as the Cores are supported by the Biomedical Research Program at Weill Cornell Medical College in Qatar (BMRP), a program funded by Qatar Foundation. Additional funding was provided by NPRP-Standard (NPRP-S) 13th Cycle grant 13S-0206-200274 from the Qatar National Research Fund (a member of Qatar Foundation). The findings herein reflect the work and are solely the responsibility of the authors. This work was also supported by NIH R01AR076029 and NIH R21ES032347 to QC.

Supplemental figures and tables

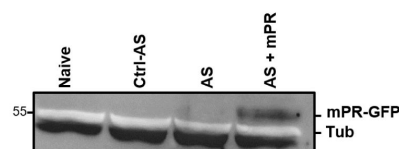
A RNA seq data (TPM) from Xenbase

Gene	Ooc S.	Ooc L.	Egg S.	Egg L.	Ovary S.	Ovary L.
PAQR5 (mPR γ)	7.07	14	1.54	1.17	4.8	19.05
PAQR6 (mPR δ)	0.26	0.07	0.13	0	0.4	0.4
PAQR7 (mPR α)	NE	NE	NE	NE	NE	NE
PAQR8 (mPR β)	0.02	1.9	0.81	0	0.05	1.83
PAQR9 (mPR ϵ)	0	0	0	0	0	0
ABHD2	1.31	1.4	3.22	3.73	4.36	21.78
nuclear P4 receptor PGR	0.06	0.02	0.22	0	0.25	0.25

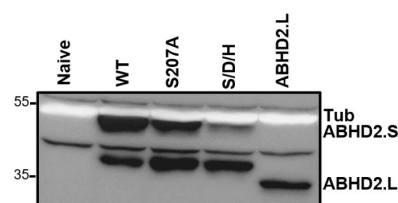
B



C

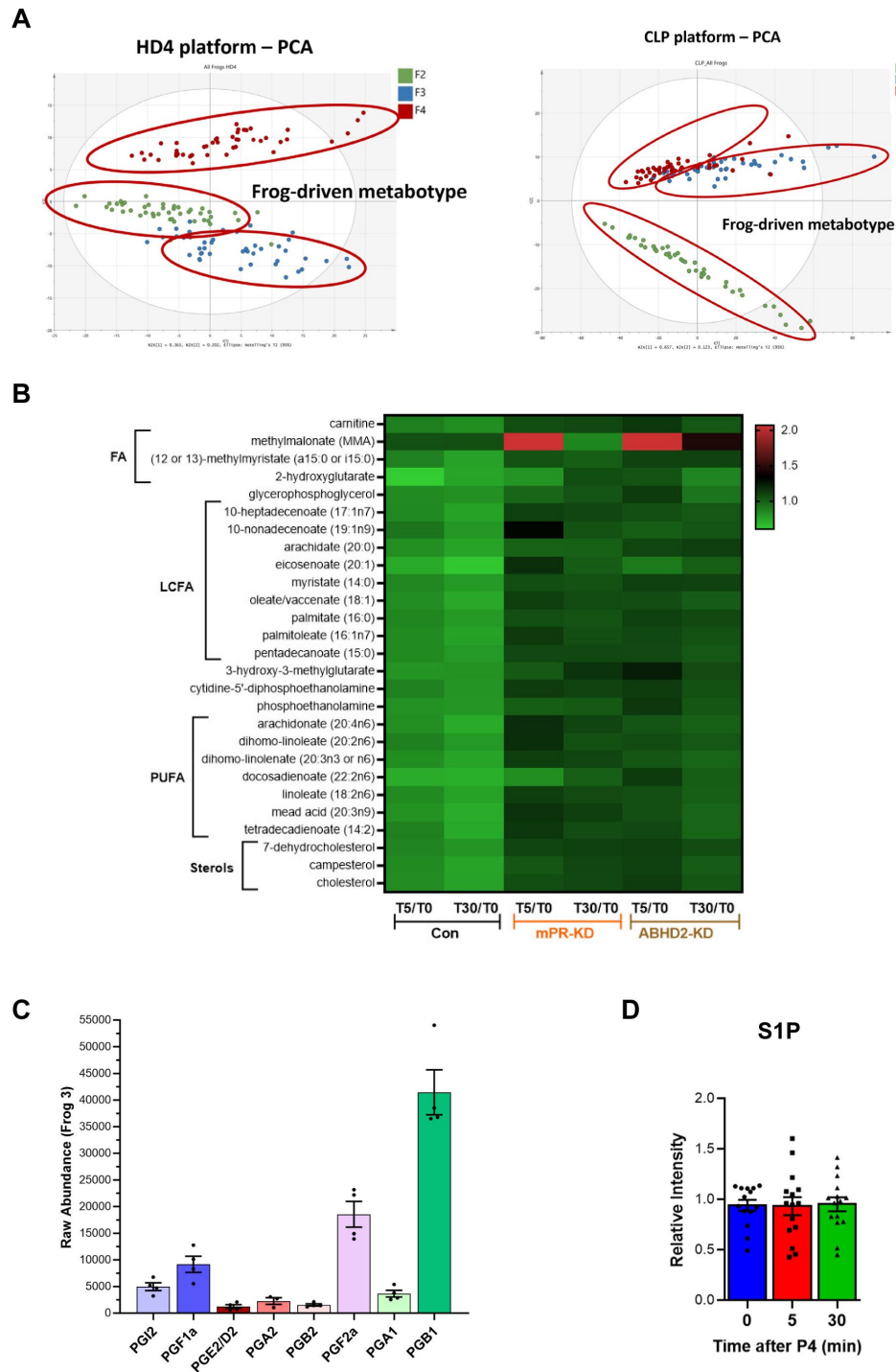


D



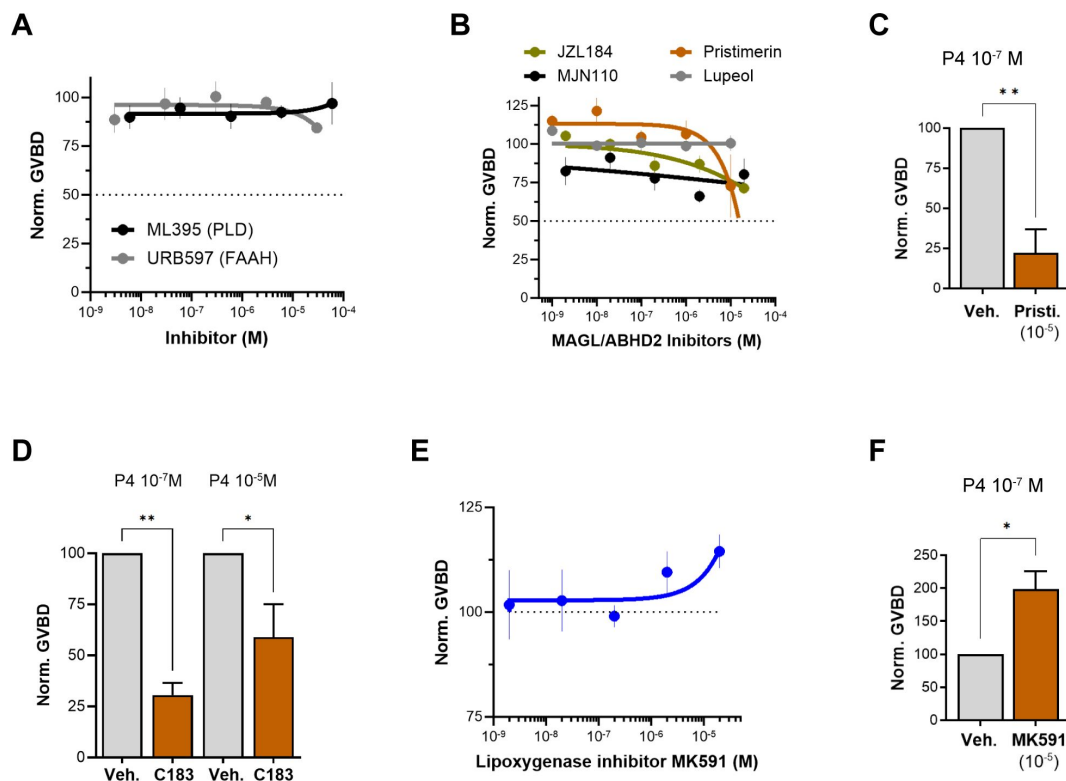
Supplemental Figure 1

A. Table representing the number of transcripts per million (TPM) of the different progesterone receptors in *Xenopus laevis* oocytes (Ooc. S., and Ooc. L.), eggs (Egg S., and Egg L.) and the ovary. Being tetraploid, *Xenopus laevis* expresses two different versions of each gene; S and L. the data are retrieved from Xenbase. **B.** Alignment of *Xenopus laevis* ABHD2.L and ABHD2.S amino acids sequences with human ABHD2. Residues implicated in the lipase activity of ABHD2 are highlighted in green, whereas the acetyltransferase domain is highlighted in purple. **C.** Representative WB of mPR-GFP proteins expression in naive, control antisense (Ctrl AS), ABHD2 antisense (AS), and AS+mPR-GFP injected oocytes. Tubulin is shown as a loading control. **D.** Representative WB of ABHD2.S protein expression levels in naive oocytes and in oocytes injected with mRNA to overexpress ABHD2 wildtype (WT), ABHD2.S S207A, ABHD2.S S207A/D345A/H376A (S/D/H). Tubulin is shown as a loading control.



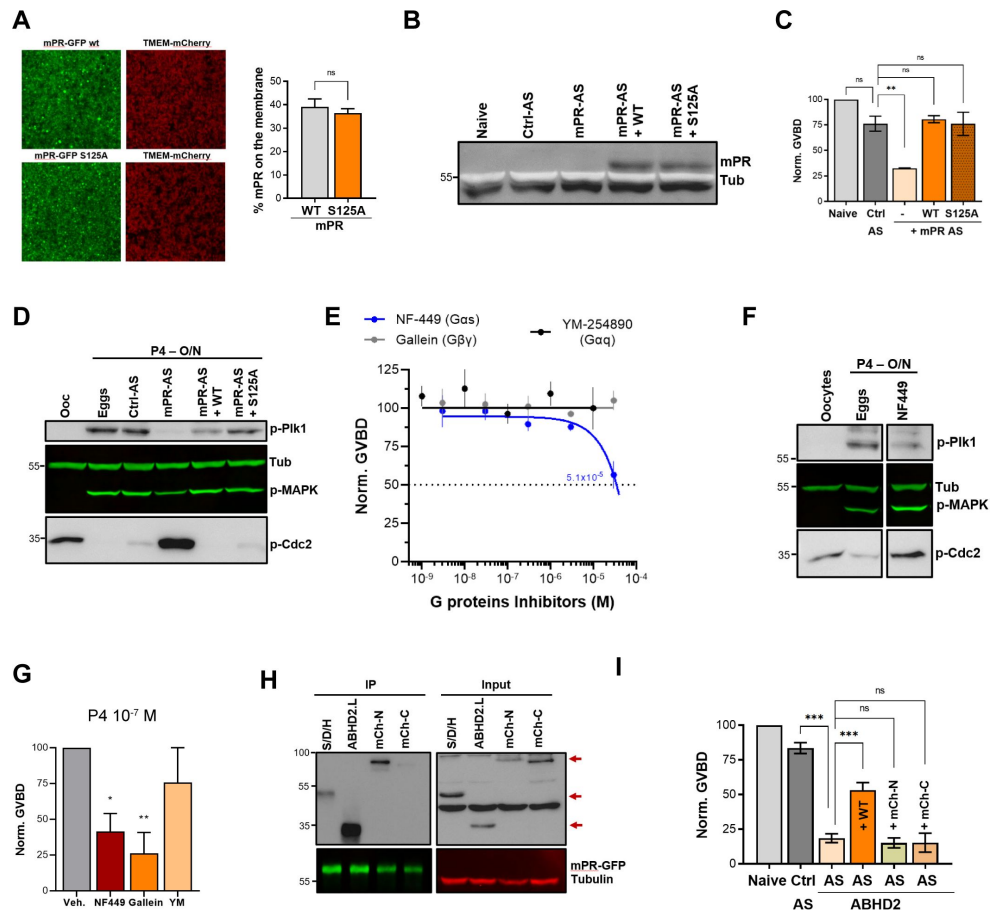
Supplemental Figure 2

A. Frog-driven metabolotype as shown from the HD4 (*left panel*) and CLP platforms (*right panel*). **B.** Data generated from the HD4 platform illustrated by the heatmap data of fold changes for individual metabolites that were changed significantly ($p < 0.05$) at either the 5 min (T5) or 30 min (T30) time points in response to P4 as compared to untreated oocytes (T5/T0 and T30/T0) for naive (Con), mPR-KD, and ABHD2-KD oocytes. Metabolites are clustered at the levels of fatty acids (FA), long chain fatty acids (LCFA), polyunsaturated fatty acids (PUFA), and sterols. **C.** The raw abundance of individual prostaglandins species in pooled oocytes before treatment from a representative frog (#3) to illustrate relative abundance among the different PG species. **D.** S1P measurements in pooled oocytes before treatment, and at 5 and 30 min after progesterone addition. The data are shown as relative intensity normalized to the average of before treatment levels.



Supplemental Figure 3

A/B/E. Oocyte maturation inhibition dose response. Oocytes were pre-treated for 2 hours with vehicle or with increasing concentrations of specified inhibitor, followed by overnight treatment with P4 at 10^{-5} M. Oocyte maturation normalized to vehicle P4-treated oocytes. IC50 of each chemical compound was calculated using a nonlinear regression fit (mean $\pm$ SEM; n = 3 independent female frogs for each chemical compound experiment). **C/D/F.** Effect of pristimerin (**C**), compound 183 (10^{-4} M) (**D**), and MK591 (**F**) on oocytes maturation at the indicated P4 concentration. Oocytes were pre-treated for 2 hours with the vehicle or indicated inhibitor, followed by overnight treatment with P4. Oocyte maturation was normalized to the control oocytes condition (treated with vehicle) (mean $\pm$ SEM; n = 3 independent female frogs for each chemical compound experiment, unpaired t-test).



Supplemental Figure 4

A. mPR S125 plasma membrane localization. Oocytes were injected with mPR-GFP wild type (WT) or the S125A mutant, along with TMEM-mCherry as a plasma membrane (PM) marker. *Left panel*, Representative confocal image of oocytes overexpressing mPR-GFP WT or mPR-GFP S125A mutant along with TMEM-mCherry. *Right panel*, histogram showing the percentage of mPR-GFP WT or mPR-GFP S125A at the PM (mean $\pm$ SEM; $n = 10$ oocytes per clone from 2 independent female frogs, unpaired t-test). **B.** Representative WB of mPR-GFP protein expression in naïve, control antisense (Ctrl AS), mPR β antisense (AS), and AS+mPR-GFP WT or mPR-GFP S125A injected oocytes. Tubulin is shown as a loading control. **C.** Oocyte maturation in oocytes injected with control antisense (Ctrl AS) or mPR β antisense (AS) with or without mPR-GFP WT or mPR-GFP S125A expression, and normalized to P4-treated naïve oocytes (Naive) (mean $\pm$ SEM; $n = 3$ independent female frogs, ordinary one-way ANOVA). **D.** Representative WB of MAPK, Pik1 and Cdc2 phosphorylation from untreated oocytes, P4 matured eggs, oocytes injected with control antisense (Ctrl AS) or mPR β antisense (AS) with or without mPR-GFP WT or mPR-GFP S125A expression. Tubulin is shown as a loading control. **E.** Inhibitor dose response. Oocytes were pre-treated for 2 hours with vehicle or with increasing concentrations of specified inhibitor, followed by overnight treatment with P4 at 10^{-5} M. Oocyte maturation normalized to vehicle P4-treated oocytes. IC₅₀ was calculated using a nonlinear regression fit (mean $\pm$ SEM; $n = 3$ independent female frogs for each chemical compound experiment). **F.** Representative WB of MAPK, Pik1 and Cdc2 phosphorylation from untreated oocytes, oocytes pretreated with vehicle for 2 hours then matured overnight (O/N) by P4 at 10^{-7} M (eggs), and oocytes pretreated for 2 hours with NF449 then treated O/N with P4 10^{-7} M. Tubulin is shown as a loading control. **G.** Drug effect on oocyte maturation at low P4 concentration. Oocytes were pre-treated for 2 hours with vehicle or the specified inhibitor followed by overnight treatment with P4 at 10^{-7} M. Oocyte maturation was normalized to the control oocytes condition (treated with vehicle) (mean $\pm$ SEM; $n = 3$ independent female frogs for each chemical compound experiment, unpaired t-test). **H.** Representative WB of the immunoprecipitation (IP) of mPR-GFP protein expression from oocytes over-expressing mPR-GFP with ABHD2.L, mCherry tagged ABHD2.S at its C-terminal (mCh-C) or N-terminal (mCh-N), and ABHD2.S S/D/H. The input represents expression levels of the different clones before the IP. Tubulin is shown as a loading control. The red arrows point to ABHD2.S different clones. **I.** Oocyte maturation measured in oocytes injected with control antisense (Ctrl AS) or ABHD2 antisense (AS) with or without ABHD2.S wild type (AS+WT) or the mCherry tagged ABHD2.S clones, and normalized to P4-treated naïve oocytes (Naive) (mean $\pm$ SEM; $n = 3$ to 5 independent female frogs, ordinary one-way ANOVA).

REAGENT	Source	Catalogue #
Antibodies		
Rabbit polyclonal anti-ABHD2	Sigma-Aldrich	Cat# HPA005999
Rabbit polyclonal anti-GFP	Living Colors	Cat# 632381

Supplemental Table 8

List of antibodies, chemicals, and reagents

Mouse monoclonal anti-SNAP25	Biolegend	Cat# 836304
Rabbit polyclonal anti-p-plk1	Cell signaling	Cat# 5472
Mouse monoclonal anti-phospho-MAPK	Cell Signaling	Cat# 9106
Rabbit polyclonal anti-phospho-Cdc2	Cell Signaling	Cat# 9111
Rabbit polyclonal anti-Tubulin	Cell Signaling	Cat# 3873
Rabbit polyclonal anti-Cdc25C (Ser216)	Cell signaling	Cat# 9528
Rabbit polyclonal anti-S1PR3	Bioss	Cat# BS-7541R
Goat anti-rabbit IgG-HRP	Jackson ImmunoResearch	Cat# 111-035-144
Goat anti-rabbit IgG- IRDye® 800	Li-COR	Cat# 925-32211
Goat anti-mouse IgG- IRDye® 680	Li-COR	Cat# 925-68070
Goat anti-mouse IgG- IRDye® 800	Li-COR	Cat# 926-32210
Mouse polyclonal anti-His	Takara	Cat# 631212
Chemicals and kits		
Progesterone	Sigma-Aldrich	Cat# 200-350-9
Pitstop® 2	Abcam	Cat# ab120687
Pitstop® 2 - negative control	Abcam	Cat# ab120688
Mammalian ProteaseArrest™	G-Biosciences	Cat# 786-331
PhosphataseArrest™	G-Biosciences	Cat# 786-451
µMACS GFP Isolation Kit	Miltenyi Biotech	Cat# 130-091-125
ECL™ Prime Western Blotting Detection	GE	Cat# 45-002-401
The XL QuikChange mutagenesis kit	Agilent Technologies	Cat# 200516
The mMessage mMachine T7 kit	ThermoFisher	Cat# AM1344
Fast SYBR™ Green Master Mix	ThermoFisher	Cat# 4385616
RNeasy Mini Kit	Qiagen	Cat# 74104
Rabbit Reticulocyte Lysate	Promega	Cat# L4960
ALICE®	Sigma	Cat# AL0103200
Trancend™ tRNA	Promega	Cat# L5061
Phospholipase A2 Activity assay kit	Abcam	Cat# ab273278
FAAH	Sigma-Aldrich	Cat# 341249
MJN110	Sigma-Aldrich	Cat# SML0872
JZL184	Sigma-Aldrich	Cat# J3455
Pristimerin	Sigma-Aldrich	Cat# P0020
Lupeol	Sigma-Aldrich	Cat# L5632
ML395	Merck	Cat# 532978
cPLA2 Inh II	Sigma-Aldrich	Cat# 5305380001
ACA	Sigma-Aldrich	Cat# A8486
Bromo-enol lactone	Sigma-Aldrich	Cat# B1552
AACOCF3	Abcam	Cat# ab120350
Mastoparan	Enzo	Cat# BML-G410-0001
HA130	Tocris	Cat# 4196
MK591	Sigma-Aldrich	Cat# 147030-01-1
NS398	Sigma-Aldrich	Cat# 349254
SC-560	Sigma-Aldrich	Cat# 565610
D-erythro MAPP	Enzo	Cat# SL221
L-erythro MAPP	Enzo	Cat# SL222

Supplemental Table 8 (continued)

Supplemental Table 8 (continued)

MP-A08	Tocris	Cat# 5803
NF-449	Sigma-Aldrich	Cat# 480420
Gallein	Sigma-Aldrich	Cat# 371708
YM-254890	Cayman	Cat# 29735
Org OD-02-0	Axon Med Chem	Cat# 2085
Software		
Image Studio 5.2	LI-COR Biosciences	N/A
ImageJ	Schneider et al. 2012	N/A
Prism	GraphPad inc.	N/A
Illustrator CC	Adobe Systems Inc	N/A
Photoshop CC	Adobe Systems Inc	N/A
ZEN 2.3 black	Zeiss	N/A
Quant Studio Real-Time PCR software.Ink	ThermoFisher	N/A
Biorender	Biorender	N/A

Supplemental Table 9

List of antisense oligonucleotides and primers

Control antisense	gcgactatacgcgcaatatg
ABHD2.LS antisense	aactcccatgtgcacata
mPR antisense	cggtagtcatgtagtaggg
S1PR3 antisense	actcacaggcaacatctat
mPR S125A – F	tcacctatctcacatgcgccatcctggctcatctgc
mPR S125A - R	gcagatgagccaggatggcgcatgtgagataggtga
ABHD2.S H120A - F	ttttctctgtgcaggcttcagctaattggtcaaataggtcaag
ABHD2.S H120A - R	cttgacctatttgaaccattagctgaagcctgcacaggagaaaa
ABHD2.S N125A - F	cagattaccatgggtgacagcttctcctgtgcagtggttcag
ABHD2.S N125A - R	ctgaacactgcacaggagaagctgtcaccatggtaactctg
Xenopus mPR β – F	cccgttgctccaccggatagt
Xenopus mPR β – R	ggtgaccgtgcctataaaa
Xenopus ABHD2.LS – F	atgggagtttgagctatggta
Xenopus ABHD2.LS – R	actctctcttggttgatgca
Xenopus ABHD2.S – F	gaggaaattccacggttacaag
Xenopus ABHD2.S – R	catgtagaggcaataaccacaa
Xenopus S1PR2 – F	ggttgccggcatttatcatcc
Xenopus S1PR2– R	gtggccactttaaactctgcc
Xenopus S1PR3 – F	tgctccacaatctgcctctg
Xenopus S1PR3– R	tcctgagaagtgccatggatc
Xenopus ODC – F	gccattgtgaagactctctccattc
Xenopus ODC – R	ttcgggtgattccttgccac

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Reviewer #1 (Public review):

Summary:

Numerous pathways have been proposed to elucidate the nongenomic actions of progesterone within both male and female reproductive tissues. The authors employed the *Xenopus* oocyte system to investigate the PLA2 activity of ABHD2 and the downstream lipid mediators in conjunction with mPRb and P4, on their significance in meiosis. The research has been conducted extensively and is presented clearly.

Strengths:

While the interaction between membranous PR and ABHD2 is not a novel concept, this present study exhibits several strengths:

- (1) mPRbeta, a member of the PAQR family, has been elusive in terms of detailed signal transduction. Through mutation studies involving the Zn binding domain, the authors discovered that the hydrolase activity of mPRbeta is not essential for meiosis and oocyte maturation. Instead, they suggest that ABHD2, acting as a coreceptor of mPRbeta, demonstrates phospholipase activity, indicating that downstream lipid mediators may play a dominant role when stimulated by progesterone.
- (2) Extensive exploration of downstream signaling pathways and the identification of several potential meiotic activity-related lipid mediators make this aspect of the study novel and potentially significant.

Weaknesses:

However, there are some weaknesses and areas that need further clarification:

- (1) The mechanism governing the molecular assembly of mPRbeta and ABHD2 remains unclear. Are they constitutively associated or is their association ligand-dependent? Does P4 bind not only to mPRbeta but also to ABHD2, as indicated in Figure 6J? In the latter case, the reviewer suggests that the authors conduct a binding experiment using labeled P4 with ABHD2 to confirm this interaction and assess any potential positive or negative cooperativity with a partner receptor.
- (2) The authors have diligently determined the metabolite profile using numerous egg cells. However, the interpretation of the results appears incomplete, and inconsistencies were noted between Figure 2F and Supplementary Figure 2C. Furthermore, PGE2 and D2 serve distinct roles and have different elution patterns by LC-MS/MS, thus requiring separate measurements. In addition, the extremely short half-life of PGI2 necessitates the measurement of its stable metabolite, 6-keto-PGF1a, instead. The authors also need to clarify

why they measured PGF1a but not PGF2a. Unfortunately, even in the revision, authors did not adequately address the last issue (differential measurements of PGD2 and E2, 6-keto-PG!alpha be determined instead of PGI2).

(3) Although they propose PGs, LPA and S1P are important downstream mediators, the exact roles of the identified lipid mediators have not been clearly demonstrated, as receptor expression and activation were not demonstrated. While the authors showed S1PR3 expression and its importance by genetic manipulation, there was no observed change in S1P levels following P4 treatment (Supplementary Figure 2D). It is essential to identify which receptors (subtypes) are expressed and how downstream signaling pathways (PKA, Ca, MAPK, etc.) relate to oocyte phenotypes.

These clarifications and further experiments would enhance the overall impact and comprehensiveness of the study.

Comments on revisions:

Need correction and addition for differential analyses of PGD2 and PGE2, and measurement of 6-keto-PGF1alpha instead of PGI2 (Figure 2F). PGI2 is extremely unstable (T1/2, 1 min in neutral buffer) and rapidly converted nonenzymically to 6-keto-PGF1a.

<https://doi.org/10.7554/eLife.92635.2.sa3>

Reviewer #2 (Public review):

Summary:

This interesting paper examines the earliest steps in progesterone-induced frog oocyte maturation, an example of non-genomic steroid hormone signaling that has been studied for decades but is still very incompletely understood. In fish and frog oocytes it seems clear that mPR proteins are involved, but exactly how they relay signals is less clear. In human sperm, the lipid hydrolase ABHD2 has been identified as a receptor for progesterone, and so the authors here examine whether ABHD2 might contribute to progesterone-induced oocyte maturation as well. The main results are:

(1) Knocking down ABHD2 makes oocytes less responsive to progesterone, and ectopically expressing ABHD2.S (but not the shorter ABHD2.L gene product) partially rescues responsiveness. The rescue depends upon the presence of critical residues in the protein's conserved lipid hydrolase domain, but not upon the presence of critical residues in its acyltransferase domain.

(2) Treatment of oocytes with progesterone causes a decrease in sphingolipid and glycerophospholipid content within 5 min. This is accompanied by an increase in LPA content and arachidonic acid metabolites. These species may contribute to signaling through GPCRs. Perhaps surprisingly, there was no detectable increase in sphingosine-1-phosphate, which might have been expected given the apparent substantial hydrolysis of sphingolipids. The authors speculate that S1P is formed and contributes to signaling but diffuses away.

(3) Pharmacological inhibitors of lipid-metabolizing enzymes support, for the most part, the inferences from the lipidomics studies, although there are some puzzling findings. The puzzling findings may be due to uncertainty about whether the inhibitors are working as advertised.

(4) Pharmacological inhibitors of G-protein signaling support a role for G-proteins and GPCRs in this signaling, although again there are some puzzling findings.

(5) Reticulocyte expression supports the idea that mPR β and ABHD2 function together to generate a progesterone-regulated PLA2 activity.

(6) Knocking down or inhibiting ABHD2 inhibited progesterone-induced mPR β internalization, and knocking down ABHD2 inhibited SNAP25 Δ 20-induced maturation.

Strengths:

All in all, this could be a very interesting paper and a nice contribution. The data add a lot to our understanding of the process, and, given how ubiquitous mPR and AdipoQ receptor signaling appear to be, something like this may be happening in many other physiological contexts.

Weaknesses:

I have several suggestions for how to make the main points more convincing.

Main criticisms:

(1) The ABHD2 knockdown and rescue, presented in Fig 1, is one of the most important findings. It can and should be presented in more detail to allow the reader to understand the experiments better. E.g.: the antisense oligos hybridize to both ABHD2.S and ABHD2.L, and they knock down both (ectopically expressed) proteins. Do they hybridize to either or both of the rescue constructs? If so, wouldn't you expect that both rescue constructs would rescue the phenotype, since they both should sequester the AS oligo? Maybe I'm missing something here.

In addition, it is critical to know whether the partial rescue (Fig 1E, I, and K) is accomplished by expressing reasonable levels of the ABHD2 protein, or only by greatly overexpressing the protein. The author's antibodies do not appear to be sensitive enough to detect the endogenous levels of ABHD2.S or .L, but they do detect the overexpressed proteins (Fig 1D). The authors could thus start by microinjecting enough of the rescue mRNAs to get detectable protein levels, and then titer down, assessing how low one can go and still get rescue. And/or compare the mRNA levels achieved with the rescue construct to the endogenous mRNAs.

Finally, please make it clear what is meant by $n = 7$ or $n = 3$ for these experiments. Does $n = 7$ mean 7 independently lysed oocytes from the same frog? Or 7 groups of, say, 10 oocytes from the same frog? Or different frogs on different days? I could not tell from the figure legends, the methods, or the supplementary methods. Ideally one wants to be sure that the knockdown and rescue can be demonstrated in different batches of oocytes, and that the experimental variability is substantially smaller than the effect size.

(2) The lipidomics results should be presented more clearly. First, please drop the heat map presentations (Fig 2A-C) and instead show individual time course results, like those shown in Fig 2E, which make it easy to see the magnitude of the change and the experiment-to-experiment variability. As it stands, the lipidomics data really cannot be critically assessed.

[Even as heat map data go, panels A-C are hard to understand. The labels are too small, especially on the heat map on the right side of panel B. And the 25 rows in panel C are not defined (the legend makes me think the panel is data from 10 individual oocytes, so are the 25 rows 25 metabolites? If so, are the individual oocyte data being collapsed into an average? Doesn't that defeat the purpose of assessing individual oocytes?) And those readers with red-green colorblindness (8% of men) will not be able to tell an increase from a decrease. But please don't bother improving the heat maps; they should just be replaced with more-informative bar graphs or scatter plots.]

(3) The reticulocyte lysate co-expression data are quite important, and are both intriguing and puzzling. My impression had been that to express functional membrane proteins, one needed to add some membrane source, like microsomes, to the standard kits. Yet it seems like

co-expression of mPR and ABHD2 proteins in a standard kit is sufficient to yield progesterone-regulated PLA2 activity. I could be wrong here-I'm not a protein expression expert-but I was surprised by this result, and I think it is critical that the authors make absolutely certain that it is correct. Do you get much greater activities if microsomes are added? Are the specific activities of the putative mPR-ABHD2 complexes reasonable?

Comments on revisions:

The authors have satisfied my concerns with their response letter and revisions.

<https://doi.org/10.7554/eLife.92635.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors report two P4 receptors, ABHD2 and mPR β that function as co-receptors to induce PLA2 activity and thus drive meiosis. In their experimental studies, the authors knock down ABHD2 and demonstrated inhibition of oocyte maturation and inactivation of Plk1, MAPK, and MPF, which indicated that ABHD2 is required for P4-induced oocyte maturation. Next, they showed three residues (S207, D345, H376) in the lipase domain that are crucial for ABHD2 P4-mediated oocyte maturation in functional assays. They performed global lipidomics analysis on mPR β or ABHD2 knockdown oocytes, among which the downregulation of GPL and sphingolipid species were observed and enrichment in LPA was also detected using their metabolomics method. Furthermore, they investigated pharmacological profiles of enzymes predicted to be important for maturation based on their metabolomic analyses and ascertained the central role for PLA2 in inducing oocyte maturation downstream of P4. They showed the modulation of S1P/S1PR3 pathway on oocyte maturation and potential role for or Gas signaling and potentially G $\beta\gamma$ downstream of P4.

Strengths:

The authors make a very interesting finding that ABHD2 has PLA2 catalytic activity but only in the presence of mPR β and P4. Finally, they provided supporting data for a relationship between ABHD2/PLA2 activity and mPR β endocytosis and further downstream signaling. Collectively, this research report defines early steps in nongenomic P4 signaling, which is of broad physiological implications.

Weaknesses:

There were concerns with the pharmacological studies presented. Many of these inhibitors are used at high (double digit micromolar) concentrations that could result in non-specific pharmacological effects and the authors have provided very little data in support of target engagement and selectivity under the multiple experimental paradigms. In addition, the use of an available ABHD2 small molecule inhibitor was lacking in these studies.

Comments on revisions:

In the revised manuscript, the authors have addressed my major concerns.

<https://doi.org/10.7554/eLife.92635.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife Assessment:

"...However, the findings are reliant on high concentrations of inhibitor drugs, and mechanistic details about the molecular interaction and respective functions of ABHD2 and mPRb are incomplete."

As discussed below in the response to Reviewers the drug concentrations used span the full dose response of the active range of each drug. In cases where the drug concentrations required to block oocyte maturation were significantly higher than those reported in the literature, we considered those drugs ineffective. In terms of the molecular details of the mechanistic interaction between mPRb and ABHD2, we now provide additional data confirming their molecular interaction to produce PLA2 activity where each protein alone is insufficient. Although these new studies provide more mechanistic insights, there remains details of the ABHD2-mPR interactions that would need to be addressed in future studies which are beyond the scope of the current already extensive study.

Public Reviews:

Reviewer 1

(1) The mechanism governing the molecular assembly of mPRbeta and ABHD2 remains unclear. Are they constitutively associated or is their association ligand-dependent? Does P4 bind not only to mPRbeta but also to ABHD2, as indicated in Figure 6J? In the latter case, the reviewer suggests that the authors conduct a binding experiment using labeled P4 with ABHD2 to confirm this interaction and assess any potential positive or negative cooperativity with a partner receptor.

The co-IP experiments presented in Figure 5E argue that the two receptors are constitutively associated at rest before exposure to P4; but at low levels since addition of P4 increases the association between mPR β and ABHD2 by ~2 folds. Importantly, we know from previous work (Nader et al., 2020) and from imaging experiments in this study that mPR recycles in immature oocytes between the PM and the endosomal compartment. It is not clear at this point within which subcellular compartment the basal association of mPR and ABHD2 occurs. We have tried to elucidate this point but have not been able to generate a functional tagged ABHD2. We generated GFP-tagged ABHD2 at both the N- and C-terminus but these constructs were not functional in terms of their ability to rescue ABHD2 knockdown. This prevented us from testing the association dynamics between ABHD2 and mPR.

Regarding whether ABHD2 in the oocyte directly binds P4 or not, we had in the initial submission no data directly supporting this rather we based the cartoon in Fig. 6J on the findings from Miller et al. (Science 2016) who showed that ABHD2 in sperm binds biotinylated P4. With the use of a new expression system to produce ABHD2 in vitro (please see below) we were able to try the experiment suggested by the Reviewer. In vitro expressed ABHD2 was incubated with biotinylated P4, and binding tested on a streptavidin column. Under these conditions we could not detect any specific binding of P4 to ABHD2, however, these experiments remain somewhat preliminary and would require validation using additional approaches to conclusively test whether *Xenopus* ABHD2 binds P4 or not. The discrepancy with the Miller et al. findings could be species specific as they tested mammalian ABHD2.

(2) The authors have diligently determined the metabolite profile using numerous egg cells. However, the interpretation of the results appears incomplete, and inconsistencies were noted between Figure 2B and Supplementary Figure 2C. Furthermore, PGE2 and D2 serve distinct roles and have different elution patterns by LC-MS/MS, thus requiring separate measurements. In addition, the extremely short half-life of PGI2 necessitates the

measurement of its stable metabolite, 6-keto-PGF1 α , instead. The authors also need to clarify why they measured PGF1 α but not PGF2 α .

We believe the Reviewer meant to indicate discrepancies between Fig. 2E (not 2B) and Supp. Fig. 2C. Indeed, the Reviewer is correct, and this is because Fig. 2E shows pooled normalized data on a per PG species and frog, whereas Supp. Fig. 2E shows an example of absolute raw levels from a single frog to illustrate the relative basal abundance of the different PG species. We had failed to clarify this in the Supp. Fig. 2E figure legend, which we have now added in the revised manuscript. So, the discrepancies are due to variation between different donor animals which is highlighted in Supp. Fig. 2A. Furthermore, to minimize confusion, in the revised manuscript we revised Supp. Fig. 2C to show only PG levels at rest, to illustrate basal levels of the different PG species relative to each other, which is the goal of this supplemental figure.

(3) Although they propose PGs, LPA, and S1P are important downstream mediators, the exact roles of the identified lipid mediators have not been clearly demonstrated, as receptor expression and activation were not demonstrated. While the authors showed S1PR3 expression and its importance by genetic manipulation, there was no observed change in S1P levels following P4 treatment (Supplementary Figure 2D). It is essential to identify which receptors (subtypes) are expressed and how downstream signaling pathways (PKA, Ca, MAPK, etc.) relate to oocyte phenotypes.

We agree conceptually with the Reviewer that identifying the details of the signaling of the different GPCRs involved in oocyte maturation would be interesting. However, our lipidomic data argue that the activation of a PLA2 early in the maturation process in response to P4 leads to the production of multiple lipid messengers that would activate GPCRs and branch out the signaling pathway to activate various pathways required for the proper and timely progression of oocyte maturation. Preparing the egg for fertilization is complex; so, it is not surprising that a variety of pathways are activated simultaneously to properly initiate both cytoplasmic and nuclear maturation to transition the egg from its meiotic arrest state to be ready to support the rapid growth during early embryogenesis. We focus on the S1P signaling pathway specifically because, as pointed out by the Reviewer, we could not detect an increase in S1P even though our metabolomic data collectively argued for an increase. Our results on the S1P pathway -as well as a plethora of other studies historically in the literature that we allude to in the manuscript- argue that these different GPCRs support and regulate oocyte maturation, but they are not essential for the early maturation signaling pathway. For example, for S1P, as shown in Figure 4, the delay/inhibition of oocyte maturation due to S1PR3 knockdown can be reversed at high levels of P4, which presumably leads to higher levels of other lipid mediators that would bypass the need for signaling through S1PR3. This is reminiscent of the kinase cascade driving oocyte maturation where there is significant redundancy and feedback regulation. Therefore, analyzing each receptor subtype that may regulate the different PG species, LPA, and S1P would be a tedious and time-consuming undertaking that goes beyond the scope of the current manuscript. More importantly based on the above arguments, we suggest that findings from such an analysis, similar to the conclusions from the S1PR3 studies (Fig. 4), would show a modulatory role on oocyte maturation rather than a core requirement for the maturation process as observed with mPR and ABHD2. Thus they would provide relatively little insights into the core signaling pathway driving P4-mediated oocyte maturation.

Reviewer 2:

(1) The ABHD2 knockdown and rescue, presented in Fig 1, is one of the most important findings. It can and should be presented in more detail to allow the reader to understand the experiments better. E.g.: the antisense oligos hybridize to both ABHD2.S and ABHD2.L,

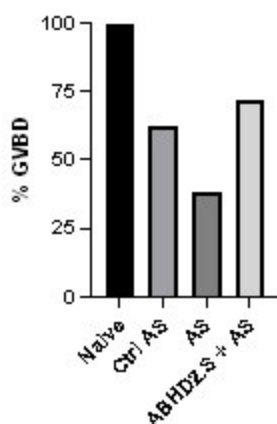
and they knock down both (ectopically expressed) proteins. Do they hybridize to either or both of the rescue constructs? If so, wouldn't you expect that both rescue constructs would rescue the phenotype since they both should sequester the AS oligo? Maybe I'm missing something here.

For the ABHD2 rescue experiment, the ABHD2 constructs (S or L) were expressed 48 hrs before the antisense was injected. The experiment was conducted in this way to avoid the potential confounding issue of both constructs sequestering the antisense. The assumption is that the injected RNA after protein expression would be degraded thus allowing the injected antisense to target endogenous ABHD2. The idea is to confirm that ABHD2.S expression alone is sufficient to rescue the antisense knockdown as confirmed experimentally.

However, to further confirm the rescue, we performed the experiment in a different chronological order, where we started with injecting the antisense to knock down endogenous ABHD2 and this was followed 24 hrs later by expressing wild type ABHD2.S. As shown in Author response image 1 this also rescues the knockdown.

Author response image 1.

ABHD2 knockdown and rescue. Oocytes were injected with control antisense (Ctrl AS) or specific ABHD2 antisense (AS) oligonucleotides and incubated at 18 oC for 24 hours. Oocytes were then injected with mRNA to overexpress ABHD.S for 48 hours and then treated with P4 overnight. The histogram shows % GVBD in naïve, oocytes injected with control or ABHD2 antisense with or without mRNA to overexpress ABHD2.S.



In addition, it is critical to know whether the partial rescue (Fig 1E, I, and K) is accomplished by expressing reasonable levels of the ABHD2 protein, or only by greatly overexpressing the protein. The author's antibodies do not appear to be sensitive enough to detect the endogenous levels of ABHD2.S or .L, but they do detect the overexpressed proteins (Fig 1D). The authors could thus start by microinjecting enough of the rescue mRNAs to get detectable protein levels, and then titer down, assessing how low one can go and still get rescue. And/or compare the mRNA levels achieved with the rescue construct to the endogenous mRNAs.

The dose response of ABHD2 protein expression in correlation with rescue of the ABHD2 knockdown is shown indirectly in Figure 1I and 1J. In experiments ABHD2 knockdown was rescued using either the WT protein or two mutants (H120A and N125A). All three constructs rescued ABHD2 KD with equal efficiency (Fig. 1I), eventhough their expression levels varied (Fig. 1J). The WT protein was expressed at significantly higher levels than both mutants, and

N125A was expressed at higher levels than H120A (Fig. 1J), note the similar tubulin loading control. Crude estimation of the WBs argues for the WT protein expression being ~3x that of H120A and ~2x that of N125A, yet all three have similar rescue of the ABHD2 knockdown (Fig. 1I). This argues that low levels of ABHD2 expression is sufficient to rescue the knockdown, consistent with the catalytic enzymatic nature of the ABHD2 PLA2 activity.

Finally, please make it clear what is meant by $n = 7$ or $n = 3$ for these experiments. Does $n = 7$ mean 7 independently lysed oocytes from the same frog? Or 7 groups of, say, 10 oocytes from the same frog? Or different frogs on different days? I could not tell from the figure legends, the methods, or the supplementary methods. Ideally one wants to be sure that the knockdown and rescue can be demonstrated in different batches of oocytes, and that the experimental variability is substantially smaller than the effect size.

The n reflects the number of independent female frogs. We have added this information to the figure legends. For each donor frog at each time point 10-30 oocytes were used.

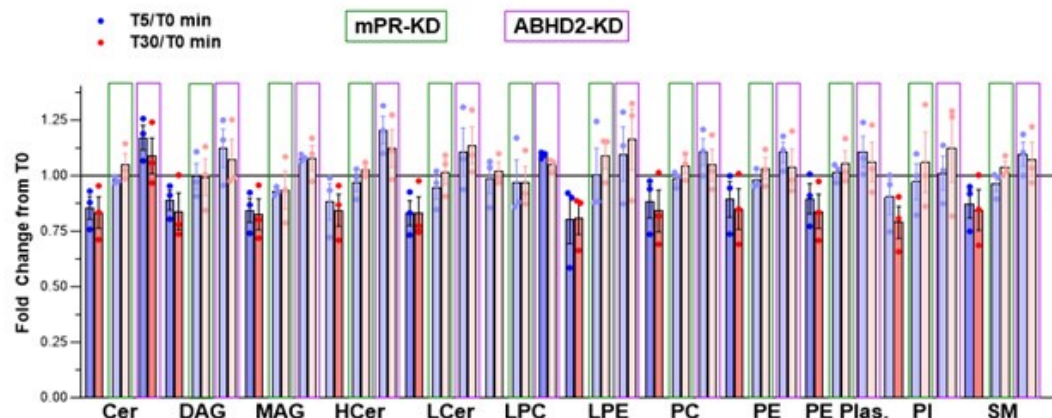
(2) The lipidomics results should be presented more clearly. First, please drop the heat map presentations (Fig 2A-C) and instead show individual time course results, like those shown in Fig 2E, which make it easy to see the magnitude of the change and the experiment-to-experiment variability. As it stands, the lipidomics data really cannot be critically assessed.

[Even as heat map data go, panels A-C are hard to understand. The labels are too small, especially on the heat map on the right side of panel B. The 25 rows in panel C are not defined (the legend makes me think the panel is data from 10 individual oocytes, so are the 25 rows 25 metabolites? If so, are the individual oocyte data being collapsed into an average? Doesn't that defeat the purpose of assessing individual oocytes?) And those readers with red-green colorblindness (8% of men) will not be able to tell an increase from a decrease. But please don't bother improving the heat maps; they should just be replaced with more informative bar graphs or scatter plots.]

We have revised the lipidomics data as requested by the Reviewer. The Reviewer asked that we show the data as a time course with each individual frog as in Fig. 2E. This turns out to be confusing and not a good way to present the data (please see Author response image 2).

Author response image 2.

Metabolite levels from 5 replicates of 10 oocytes each at each time point were measured and averaged per frog and per time point. Fold change was measured as the ratio at the 5- and 30-min time points relative to untreated oocytes (T0). FCs that are not statistically significant are shown as faded. Oocytes with mPR knockdown (KD) are boxed in green and ABHD2-KD in purple.



We therefore revised the metabolomics data as follow to improve clarity. The changes in the glycerophospholipids and sphingolipids determined on the Metabolon CLP platform (specific for lipids) are now shown as single metabolites clustered at the levels of species and pathways and arranged for the 5- and 30-min time points sequentially on the same heatmap as requested (Fig. 2B). This allows for a quick visual overview of the data that clearly shows the decrease in the lipid species following P4 treatment in the control oocytes and not in the mPR-KD or ABHD2-KD cells (Fig. 2B). The individual species are listed in Supplemental Tables 1 and 2. We also revised the Supplemental Tables to include the values for the non-significant changes, which were omitted from the previous submission.

We revised the metabolomics data from the HD4 platform in a similar fashion but because the lipid data were complimentary and less extensive than those from the CLP platform, we moved that heatmap to Supplemental Fig. 2B.

For the single oocyte metabolomics, we now show the data as the correlation between FC and p value, which clearly shows the upregulated (including LPA) and downregulated metabolites at T30 relative to T0 (Fig. 2C). The raw data is now shown in a new Supplemental Table 7.

(3) The reticulocyte lysate co-expression data are quite important and are both intriguing and puzzling. My impression had been that to express functional membrane proteins, one needed to add some membrane source, like microsomes, to the standard kits. Yet it seems like co-expression of mPR and ABHD2 proteins in a standard kit is sufficient to yield progesterone-regulated PLA2 activity. I could be wrong here - I'm not a protein expression expert - but I was surprised by this result, and I think it is critical that the authors make absolutely certain that it is correct. Do you get much greater activities if microsomes are added? Are the specific activities of the putative mPR-ABHD2 complexes reasonable?

We thank the Reviewer for this insightful comment. We agree that this is a critical result that would benefit from cross validation, especially given the low level of PLA2 activity detected in the reticulocyte lysate expression system. We have therefore expanded these studies using another in vitro expression system with microsomal membranes based on tobacco extracts (ALiCE@Cell-Free Protein Synthesis System, Sigma Aldrich) to enhance production and stability of the expressed receptors as suggested by the Reviewer. We further prepared virus-like particles (VLPs) from cells expressing each receptor individually or both receptors together. We however could not detect any PLA2 activity from the VLPs. We thus focused on the coupled in vitro transcription/translation tobacco extracts that allow the expression of difficult-to-produce membrane proteins in microsomes. This kit targets membrane protein directly to microsomes using a microsome targeting melittin signal peptide. This system took

significant time and effort to troubleshoot and adapt to mPR and ABHD2 expression. We were however ultimately able to produce significantly higher amounts of both ABHD2 and mPRb, which were readily detected by WBs (Supplemental Fig. 4I). In contrast, we could not reliably detect mPR or ABHD2 using WBs from reticulocyte lysates given the limited amounts produced.

Similarly to our previous findings with proteins produced in reticulocytes, expression of ABHD2 or mPR β alone was not associated with an increase in PLA2 activity over a two-hour incubation period (Fig. 5C). It is worth noting here that the tobacco lysates had high endogenous PLA2 activity. However, co-expression of both mPRb and ABHD2 produced robust PLA2 activity that was significantly higher than that detected in reticulocyte lysate system (Fig. 5C). Surprisingly, however this PLA2 activity was P4 independent as it was observed when both receptors are co-expressed in the absence of P4.

These results validate our earlier conclusion that PLA2 activity requires both mPR and ABHD2, so their interaction is needed for enzymatic activity. It is interesting however that in the tobacco expression system this mPR-ABHD2 PLA2 activity becomes for the most part P4 independent. As the tobacco expression system forces both ABHD2 and mPR into microsomes using a signal sequence, the two receptors are enriched in the same vesicular compartment. As they can interact independently of P4 as shown in the co-IP experiments in immature oocytes (Fig. 5D), their forced co-expression in the same microsomal compartment could lead to their association and thus PLA2 activity. This is an attractive possibility that fits the current data, but would need independent validation.

Reviewer 3:

There were concerns with the pharmacological studies presented. Many of these inhibitors are used at high (double-digit micromolar) concentrations that could result in non-specific pharmacological effects and the authors have provided very little data in support of target engagement and selectivity under the multiple experimental paradigms. In addition, the use of an available ABHD2 small molecule inhibitor was lacking in these studies.

For the inhibitors used we performed a full dose response to define the active concentrations. So, inhibitors were not used at one high dose. We then compared the EC50 for each active inhibitor to the reported EC50 in the literature (Table 1). The inhibitors were deemed effective only if they inhibited oocyte maturation within the range reported in the literature. This despite the fact that frog oocytes are notorious in requiring higher concentrations of drug given their high lipophilic yolk content, which acts as a sponge for drugs. So our criteria for an effective inhibitor are rather stringent.

Based on these criteria, only 3 inhibitors were 'effective' in inhibiting oocyte maturation: Ibuprofen, ACA and MP-A08 with relative IC50s to those reported in the literature of 0.7, 1.1, and 1.6 respectively. Ibuprofen targets Cox enzymes, which produce prostaglandins. We independently confirmed an increase in PGs in response to P4 in oocytes thus validating the drug inhibitory effect. ACA blocks PLA2 and inhibits maturation, a role supported by the metabolomics analyses that shows decrease in the PE/PE/LPE/LPC species; and by the ABHD2-mPR PLA2 activity following in vitro expression. Finally, MP-A08 blocks sphingosine kinase activity, which role is supported by the metabolomics showing a decrease in sphingosine levels in response to P4; and our functional studies validating a role for the S1P receptor 3 in oocyte maturation.

As pointed out by the Reviewer, other inhibitors did block maturation at very high concentration, but we do not consider these as effective and have not implicated the blocked enzymes in the early steps of oocyte maturation. To clarify this point, we edited the summary panel (now Fig. 2D) to simplify it and highlight the inhibitors with an effect in the reported

range in red and those that don't inhibit based on the above criteria in grey. Those with intermediate effects are shown in pink. We hope these edits clarify the inhibitors studies.

Recommendations For the Authors

Reviewer 2:

(1) Introduction, para 1. Please change "mPRs mediated" to "mPR-mediated".

Done

(2) Introduction, para 2. Please change "cyclin b" to "cyclin B".

Done

(3) Introduction, para 2. Please change "that serves" to "which serves".

Done

(4) Introduction, para 4. I know that the authors have published evidence that "a global decrease in cAMP levels is not detectable" (2016), but old work from Maller and Krebs (JBC 1979) did see an early, transient decrease after P4 treatment, and subsequent work from Maller said that there was both a decrease in adenylyl cyclase activity and an increase in cAMP activity. Perhaps it would be better to say something like "early work showed a transitory drop in cAMP activity within 1 min of P4 treatment (Maller), although later studies failed to detect this drop and showed that P4-dependent maturation proceeds even when cAMP is high (25)".

We agree and thank the Reviewer for this recommendation. The text was revised accordingly.

(5) Results, para 1. Based on the results in Fig 1B, one should probably not assert that ABHD2 is expressed "at levels similar to those of mPR β in the oocyte"-with different mRNAs and different PCR primers, it's hard to say whether they are similar or not. The RNAseq data from Xenbase in Supp Fig 1 supports the idea that the ABHD2 and mPR β mRNAs are expressed at similar levels at the message level, although of course mRNA levels and protein levels do not correlate well when different gene products are compared (Wuhr's 2014 Curr Biol paper reported correlation coefficients of about 0.3).

We agree and have changed the text as follow to specifically point out to RNA: "we confirmed that ABHD2 RNA is expressed in the oocyte at levels similar to those of mPR β RNA (Fig. 1B)."

(6) Results, para 2. It would be worth pointing out that since an 18 h incubation with microinjected antisense oligos was sufficient to substantially knock down both the ABHD2 mRNAs (Fig 1C) and the ectopically-expressed proteins (Fig 1D), the mRNA and protein half-lives must be fairly short, on the order of a few hours or less.

Done

(7) Figure 1. Please make the western blots (especially Fig 1D) and their labeling larger. These are key results and as it stands the labeling is virtually unreadable on printed copies of the figures. I'm not sure about eLife's policy, but many journals want the text in figures to be no smaller than 5-7 points at 100% size.

Likewise for many of the western blots in subsequent figures.

As requested by the Reviewer we have increased the font and size of all Western blots in the Figures.

(8) Figure 1E, G. I am not sure one should compare the effectiveness of the ABHD2 rescue (Fig 1E) and the mPR β rescue (Fig 1G). Even if these were oocytes from the same frog, we do not know how the levels of the overexpressed ABHD2 and mPR β proteins compare. E.g. maybe ABHD2 was highly overexpressed and mPR β was overexpressed by a tiny amount.

Although this is a possibility, the expression levels of the proteins here is not of much concern because we previously showed that mPR β expression effectively rescues mPR β antisense knockdown which inhibits maturation (please see (Nader et al., 2020)). This argues that at the levels of mRNA injected mPR is functional to support maturation, yet it does not rescue ABHD2 knockdown to the same levels (Fig. 1G). With that it is fair to argue that mPR β is not as effective at rescuing ABHD2 KD maturation.

(9) Inhibitor studies: There are two likely problems in comparing the observed potencies with legacy data - in vitro vs in vivo data and frog vs. mammalian data. Please make it clear what is being compared to what when you are comparing legacy data.

The legacy data are from the literature based on the early studies that defined the IC₅₀ for inhibition primarily using in vivo models (cell line mostly) but not oocytes. Typically, frog oocytes require significantly higher concentrations of inhibitors to mediate their effect because of the high lipophilic yolk content which acts as a sponge for some drugs. So, the fact that the drugs that are effective in inhibiting oocyte maturation (ACA, MP-A08, and Ibuprofen) work in a similar or lower concentration range to the published IC₅₀ gives us confidence as to the specificity of their effect. We have revised Table 1 to include the reference for each IC₅₀ value from the literature to allow the reader to judge the exact model and context used.

(10) Isn't it surprising that Gas seems to promote maturation, given the Maller data (and data from others) that cAMP and PKA oppose maturation (see also the authors' own Fig 1A) and the authors' previous data sees no positive effect (minor point 7 above)?

We show that a specific Gas inhibitor NF-449 inhibits maturation (although at relatively high concentrations), which is consistent with a positive role for Gas in oocyte maturation. We argue based on the lipidomics data and the inhibitors data that GPCRs play a modulatory role and not a central early signaling role in terms of releasing oocyte meiotic arrest. They are likely to have effects on the full maturation of the egg in preparation for embryonic development. The actions of the multiple lipid messengers generated downstream of mPR β activation are likely to act through GPCRs and could signal through Gas or other Ga or even through G β y. Minor point 7 refers to the size of Western blots.

(11) Page 9, bottom: "...one would predict activation of sphingosine kinases...." Couldn't it just be the activity of some constitutively active sphingosine kinase? Maybe replace "activation" with "activity".

A constitutively sphingosine kinase activity would not make sense as it needs to be activated by P4.

(12) Sometimes the authors refer to concentrations in molar units plus a power of 10 (e.g. 10⁻⁵ M) and sometime in μ M or nM, sometimes even within the same paragraph. This makes it unnecessarily difficult to compare. Please keep consistent.

We replaced all the concentrations through the text to M with scientific notation for consistency as requested by the Reviewer.

| (13) Fig 3I: "Sphingosine kinase" is misspelled.

This has been corrected. We thank the Reviewer for catching it.

| (14) Legend to Fig. 5: Please change "after P4 treatment in reticulocytes" to "after P4 treatment in reticulocyte lysates".

Done

| (15) Fig 6J. Doesn't the MAPK cascade inhibit MYT1? I.e. shouldn't the arrow be -| rather than ->?

Yes the Reviewer is correct. This has been changed. We thank the Reviewer for noticing this error.

| (16) Materials and Methods, second paragraph. Please change "inhibitor's studies" to "inhibitor studies".

Corrected thanks.

| (17) Table 1: Please be consistent in how you write Cox-2.

Done.

Reviewer #3:

The findings are of potential broad interest, but I have some concerns with the pharmacological studies presented. Many of these inhibitors are used at high (double-digit micromolar) concentrations that could result in non-specific pharmacological effects and the authors have provided very little data in support of target engagement and selectivity under the multiple experimental paradigms. Importantly, several claims regarding lipid metabolism signaling in the context of oocyte maturation are made without critical validation that the intended target is inactivated with reasonable selectivity across the proteome. Several of the inhibitors used for pharmacology and metabolomics are known covalent inhibitors (JZL184 and MJN110) that can readily bind additional lipases depending on the treatment time and concentration.

I did not find any data using the reported ABHD2 inhibitor (compound 183; PMID: 31525885). Is there a reason not to include this compound to complement the knockdown studies? I believe this is an important control given that not all lipid effects were reversed with ABHD2 knockdown. The proper target engagement and selectivity studies should be performed with this ABHD2 inhibitor.

We obtained aliquots the reported ABHD2 inhibitor compound 183 from Dr. Van Der Stelt and tested its effect on oocyte maturation at 10^{-4} M using both low (10^{-7} M) or high (10^{-5} M) P4 concentration. Compound 183 partially inhibited P4-mediated oocyte maturation. The new data was added to the manuscript as Supplemental Figure 3D.

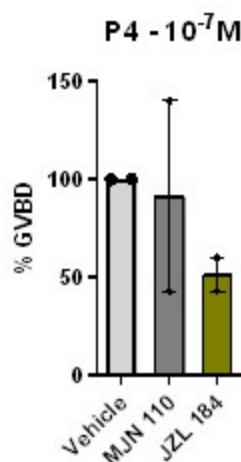
| Additional comments:

(1) *Pristimerin was tested at low P4 concentration for effects on oocyte maturation. Authors should also test JZL184 and MJN110 under this experimental paradigm.*

We have tested the effect of high concentration (2.10^{-5} M) of JZL184 or MJN110 on oocyte maturation at low P4 concentration (Author response image 3). MJN 110 did not have a prominent effect on oocyte maturation at low P4, whereas JZL184 inhibited maturation by 50%. However, this inhibition of maturation required concentrations of JZL 184 that are 10 times higher than those reported in rat and human cells (Cui et al., 2016; Smith et al., 2015), arguing against an important role for a monoacylglycerol enzymatic activity in inducing oocyte maturation.

Author response image 3.

The effect of MJN110 and JZL184 compounds on oocyte maturation at low P4 concentration. Oocytes were pre-treated for 2 hours with the vehicle or with the highest concentration of 2.10^{-5} M for both JZL184 or MJN110, followed by overnight treatment with P4 at 10^{-7} M. Oocyte maturation was measured as % GVBD normalized to control oocytes (treated with vehicle) (mean + SEM; n = 2 independent female frogs for each compound).



1. *Figure 4A showed different ct values of ODC between Oocytes and spleen, please explain them in the text. There is not any description regarding spleen information in Figure 4A, please make it clear in the text.*

We thank the Reviewer for this recommendation. The text was revised accordingly.

(3) *For Figures 3A, E, and I, there are different concentration settings for comparing the activity, is it possible to get the curves based on the same set of concentrations? The concentration gradient didn't include higher concentration points in these figures, thus the related values are incorrect. Please set more concentration points to improve the figures. And for the error bar, there are different display formats like Figure 4c and 4d, etc. Please uniform the format for all the figures. Additionally, for the ctrl. or veh., please add an error bar for all figures.*

Some of the drugs tested were toxic to oocytes at high concentrations so the dose response was adjusted accordingly. The graphs were plotted to encompass the entire tested dose

response. We could have plotted the data on the same x-axis range but that would make the figures uneven and awkward.

We are not clear what the Reviewer means by “The concentration gradient didn't include higher concentration points in these figures, thus the related values are incorrect.”

The error bars for all dose responses are consistent throughout all the Figures. They are different from those on bar graphs to improve clarity. If the Reviewer wishes to have the error bars on the bar graphs and dose response the same, we are happy to do so.

For the inhibitor studies the data were normalized on a per frog basis to control for variability in the maturation rate in response to P4, which varies from frog to frog. It is thus not possible to add error bars for the controls.

(4) Please check the sentence "However, the concentration of HA130..... higher than....."; Change "IC50" to "IC₅₀" in the text and tables. Table 1 lists IC₅₀ values in the literature, but the references are not cited. Please include the references properly. For the IC₅₀ value obtained in the research, please include the standard deviation in the table. For reference parts, Ref 1, 27, 32, 46, doublecheck the title format.

We edited the sentence as follows to be more clear: “However, this inhibition of maturation required high concentrations of HA130 -at least 3 orders of magnitude higher than the reported HA130 IC₅₀...”

We changed IC₅₀ to subscript in Table 1.

We added the relevant references in Table 1 to provide context for the cited IC₅₀ values for the different inhibitors used.

We added SEM to the IC₅₀ for inhibition of oocyte maturation values in Table 1.

We checked the titles on the mentioned references and cannot identify any problems.

References

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<https://doi.org/10.7554/eLife.92635.2.sa0>